

Paleoceanography and Paleoclimatology®



RESEARCH ARTICLE

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Key Points:

- Paleogeography and CO₂ both impact Miocene ocean oxygen and should be considered when interpreting proxy records
- High CO₂ drives Arctic and Atlantic hypoxia, intensified under MioMIP1 paleogeography where the AMOC collapses
- The Eastern Tropical Pacific OMZ contracts under the same paleogeography at high CO₂, consistent with middle Miocene proxy evidence

Supporting Information:

Supporting Information may be found in the online version of this article.

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Simulated Ocean Oxygen Under Miocene Boundary Conditions

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Abstract The Miocene Climatic Optimum (MCO; 16.9–14.7 Ma) was a warm interval with atmospheric CO₂ comparable to end-of-21st-century projections under intermediate emission scenarios. Proxy records indicate the Oxygen Minimum Zone (OMZ) in the Eastern Tropical Pacific (ETP) was small or non-existent during the MCO, expanding only when CO₂ declined after 15 Ma. Conversely, the modern ETP OMZ is expanding under anthropogenic warming. The mechanisms behind these opposing trends remain unclear. Here, we use the Geophysical Fluid Dynamics Laboratory coupled model (GFDL CM2.1) with the offline biogeochemistry PCO2 model to simulate Miocene ocean oxygen across multiple CO₂ levels and two paleogeographic configurations. Increasing CO₂ produces widespread oxygen loss at shallow and intermediate depths. One configuration develops pervasive hypoxia throughout the water column in the Atlantic and Arctic, driven by stagnation and stratification associated with a collapsed Atlantic Meridional Overturning Circulation (AMOC). In the same configuration, the ETP OMZ contracts at higher CO₂ due to weaker trade winds, reduced upwelling, and water mass transport changes through the open Central American Seaway (CAS) associated with the shutdown of North Atlantic Deep Water and Tethys Intermediate Water formation. The same circulation reorganization responsible for Atlantic deoxygenation thus contributes to ETP oxygenation, highlighting the spatial complexity of the marine oxygen response to warming. These results indicate Miocene ETP oxygen dynamics are not a perfect analog to future changes: the boundary conditions permitting ETP OMZ contraction (including an open CAS and collapsed AMOC) are absent from the modern ocean, limiting direct extrapolation from proxy evidence.

Plain Language Summary Oxygen is necessary for the function of marine ecosystems but is currently declining across the global ocean. In addition, regions of extremely low oxygen, called Oxygen Minimum Zones (OMZs), are also expanding. OMZ expansion has consequences for marine life, fisheries, and the global climate, and future changes to OMZs are uncertain. Paleoclimate data can help our understanding by showing how past warming affected ocean oxygen. Evidence from the middle Miocene (16.9–14.7 million years ago) indicates that, unlike today, the OMZ in the Eastern Pacific may have been smaller under warmer conditions. Here, we model middle Miocene oxygen to assess whether the dynamics driving this past contraction are relevant to modern warming. We find some dynamics that may play a role in the modern climate, such as weaker winds in the tropics. Other processes depend directly on the existence of ocean gateways, such as the Central American Seaway, which are closed today. We show that the Pacific OMZ contraction is strongly connected to widespread oxygen loss in the North Atlantic.

1. Introduction

Dissolved ocean oxygen (O₂) has decreased globally by over 2% since 1960 (Schmidtko et al., 2017), and is projected to continue to decrease by a further 1%–7% over the next century (Keeling et al., 2010; Long et al., 2016). Oxygen Minimum Zones (OMZs), areas of the water column with persistent low dissolved oxygen concentrations, are also expanding under modern warming (e.g., Stramma et al., 2008; Stramma et al., 2010). The combination of rising temperatures and expanding OMZs is expected to restrict the habitat of marine organisms (Deutsch et al., 2015, 2024), and impact the societies that rely on ecosystem services for sustenance and economic activity (Limburg et al., 2017, 2020).

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While 21st-century projections generally indicate further deoxygenation (Bopp et al., 2013; Kwiatkowski et al., 2020), longer-term outcomes remain uncertain, with simulated changes in ocean oxygen concentrations diverging in both magnitude and sign. Using an offline biogeochemistry model (PCO2, first published in Pasquier et al., 2023, also used in this study), Pasquier et al. (2024) simulated steady-state marine oxygen concentrations in a perpetually warmer idealized ocean built from end-of-century climate simulations, and found widespread deoxygenation throughout the entire water column, including in OMZs. In contrast, Yamamoto et al. (2015) and Frölicher et al. (2020) conducted millennial-scale transient simulations in which atmospheric CO₂ was increased to 2× or 4× pre-industrial values, and found enhanced deep-ocean oxygenation. In their simulations, global warming initially causes the expected stratification, and oxygen levels decline. However, a reinvigorated overturning returns after about 700 years, ventilating the deep ocean and leading to higher oxygenation below 2,000 m, although oxygen is found to decrease between 200 and 1,000 m owing to shallower but intensified respiration. In addition, Fu et al. (2018) simulated future warming under a Representative Concentration Pathway (RCP) 8.5 scenario and found that the current tropical OMZ expansion may eventually reverse as early as 2300, due to reductions in tropical biological export production and ventilation changes. Furthermore, outputs from the 6th phase of the Coupled Model Intercomparison Project (CMIP6) revealed additional complexity; while low-oxygen Pacific waters ($\leq 120 \mu\text{mol kg}^{-1}$) are generally expected to expand, the OMZ core ($\leq 20 \mu\text{mol kg}^{-1}$) may instead contract (Busecke et al., 2022). These studies highlight how the complex interplay between climate, ocean circulation, and biology complicates our projections of long-term ocean oxygenation. Paleoclimate simulations offer an opportunity to investigate past changes to oxygen under long-term warming and may therefore improve this understanding.

The middle Miocene Climatic Optimum (MCO, 16.9–14.7 million years ago; Ma), with atmospheric CO₂ concentrations of 450–800 ppm (CenCO₂PIP Consortium, 2023; Rae et al., 2021), offers a potential analog for future climate change within the mid-range of the future warming scenarios presented by the Intergovernmental Panel on Climate Change (IPCC), such as Shared Socioeconomic Pathways SSP2-4.5 or SSP4-6.0 (IPCC, 2023; Steinthorsdottir et al., 2021). Proxy reconstructions and paleoclimate modeling of the Miocene therefore offer an opportunity to investigate and better constrain potential future long-term climate changes. However, paleoclimate modeling studies of the Miocene have typically omitted biogeochemistry and instead been focused on physical processes (Burls et al., 2021; Micheels et al., 2007; Steinthorsdottir et al., 2021; Wei et al., 2023). While some, like Butzin et al. (2011), used carbon-cycle modeling to validate ocean model results against proxy $\delta^{13}\text{C}$ records, they did not investigate biogeochemistry specifically. A recent study using an intermediate complexity earth system model (cGENIE; Burke et al., 2026) suggested that Miocene paleogeography can enhance Atlantic and diminish Pacific Ocean oxygen depletion at higher CO₂. However, these effects remain largely unexplored in fully dynamic ocean-atmosphere models. Addressing the deficit in Miocene-era biogeochemistry modeling is important for more accurate future projections of ocean deoxygenation and its impact on ecosystems.

Reconstructing past oxygen patterns is difficult, since interpretations of oxygen proxies are often only qualitative, and comprehensive reconstructions of global oxygen are hindered by the sparsity of observations (Hoogakker et al., 2025). But some inferences can be made, such as the emergence of widespread deoxygenation during some past warm climate intervals.

There is a growing body of evidence pertaining to deoxygenation and OMZ expansion during Cenozoic warm periods. Proxy evidence from Zhou et al. (2016) suggests OMZ expansion at intermediate depths (1,500–2,000 m) in the Atlantic, Southern, and Indian Oceans during the greenhouse climates of the late Paleocene (66–56 Ma) and early Eocene (56–33.9 Ma). During this epoch, abrupt and extreme warming events such as the Paleocene-Eocene Thermal Maximum (PETM, ~56 Ma) also perturbed ocean oxygen, with proxy and model evidence of seafloor anoxia expanding on a global scale (Clarkson et al., 2021). Yao et al. (2018) modeled PETM biogeochemistry and found that widespread warming-driven oxygen decline may have led to carbon-cycle perturbations with consequences for multicellular life. Meissner et al. (2014) conducted an ensemble of late Paleocene and PETM simulations with a range of CO₂ concentrations. They find widespread tropical hypoxia, with the Arctic and Southern Ocean experiencing regional hypoxia at higher atmospheric CO₂ concentrations. This hypoxia concentrates in the upper 1,000 m for most of the ocean, but the Arctic Ocean and Atlantic Ocean experience bottom-water deoxygenation in their most carbon-intensive PETM simulation. Relevant to our study of Miocene oxygen, Zhang et al. (2025) used iron proxies to illustrate enhanced deep oxygenation during the cooler Middle Miocene Climate Transition (MMCT; ~14.7–13.8 Ma) relative to the earlier, warm MCO.

Regional evidence from other past climate episodes runs counter to this pattern. Combined foraminiferal $\delta^{15}\text{N}$ and ocean-model data for the PETM indicate contraction of the tropical North Pacific oxygen deficient zone, driven by regional declines in biological productivity (Moretti et al., 2024). Similarly, Auderset et al. (2022) utilized foraminifera-bound $\delta^{15}\text{N}$ data to demonstrate diminished water-column denitrification in both the Atlantic and Pacific during the MCO and Early Eocene Climatic Optimum. The authors attribute a strong decrease in denitrification in the Pacific to a contraction of Pacific OMZs during these two warm periods. Supplementing this conclusion for the MCO, Hess et al. (2023) found, based on combined I/Ca and $\delta^{15}\text{N}$ proxies, contraction of the Eastern Tropical Pacific (ETP) OMZ during the high- CO_2 MCO and expansion across the subsequent CO_2 decline of the MMCT. A similar, albeit smaller, contraction of the Arabian Sea OMZ has been found by Hess et al. (2026).

These findings suggest that OMZ expansion on long timescales is not a necessary consequence of higher atmospheric CO_2 concentrations. Moretti et al. (2024) used a model to attribute OMZ contraction to a decrease in primary production and nutrient availability; however, to date, no model simulations explain the increased oxygen during the warm MCO. Our model simulations seek to identify the three-dimensional mechanisms that caused the oxygen responses recorded in proxies and to assess their relevance to modern warming.

When modeling past climates, we face several uncertainties that can significantly influence results. Particularly relevant to our study is the uncertainty in paleogeography during the Miocene, as past continental arrangements and bathymetry can be reconstructed with different approaches. The Miocene Model Intercomparison Project phase 1 (MioMIP1; Burls et al., 2021) offers a standardized paleogeography based on an updated version of the Herold et al. (2008, 2011) paleogeography, but there are several other Miocene reconstructions available (Frigola et al., 2018; Scotese & Wright, 2018; Straume et al., 2020), which possess different topographic features and ocean gateways. Different continental configurations can strongly influence ocean circulation, ventilation, and thereby nutrient and oxygen distributions (e.g., Abelson & Erez, 2017; Borrelli et al., 2014; Hutchinson et al., 2019).

This study presents new biogeochemical simulations to shed light on the equilibrium response of ocean oxygen under Miocene boundary conditions to a range of atmospheric CO_2 concentrations. We also evaluate the impact of the two different paleogeography reconstructions of Straume et al. (2020) and Burls et al. (2021) on global biogeochemistry and compare our simulations to available proxy data. Finally, this study presents a plausible series of mechanisms explaining the anomalous decrease in OMZ magnitude at high CO_2 described by Auderset et al. (2022) and Hess et al. (2023).

2. Methods

2.1. Miocene GFDL CM2.1 Simulations

The climate simulations were conducted with the Geophysical Fluid Dynamics Laboratory coupled model (GFDL CM2.1; Delworth et al., 2006) using Miocene (15 Ma) boundary conditions as described by Hutchinson et al. (2025a). Model outputs pertaining to global temperature and large-scale circulation differences have been published in Hutchinson et al. (2025a). The model incorporates the Modular Ocean Model version 5.1 (MOM5; Griffies, 2012) and the GFDL Atmosphere Model 2 (AM2; GFDL Global Atmospheric Model Development Team, 2004), with resolutions of $1^\circ \times 1.5^\circ \times 50$ levels and $3^\circ \times 3.75^\circ \times 24$ levels, in the ocean and atmosphere, respectively. The ocean model uses a tripolar grid with regular latitude-longitude from 85°S to 65°N and a bipolar grid north of 65°N (Hutchinson et al., 2018).

We carried out a suite of simulations (Table 1) designed to address three major uncertainties of the Miocene, namely topography, CO_2 concentrations, and ocean circulation:

1. To address the uncertainty in continental arrangement, we embedded our model in two different configurations, based on reconstructions by Burls et al. (2021) and Straume et al. (2020), denoted by “B21” and “S20” hereafter. These reconstructions have broadly similar continental positions, but many small differences in coastlines and ocean gateways (see Figure 1), which can affect ocean circulation.
2. To address the greenhouse gas concentration uncertainty, we used atmospheric CO_2 forcings ranging from $1\times$ to $3\times$ the preindustrial level (286–858 ppm).

Table 1

Experimental Design and Resulting Large-Scale Circulation

Geography configuration	CO ₂ (ppm)	Simulation name	Initial temperature	Initial salinity	Length (yrs)	AMOC ^a	PMOC ^b
B21 ^c	286	B21-1×	Idealized ^d	34.7 psu	5,000	✓	✓
	572	B21-2×	Idealized ^d	34.7 psu	5,000		✓
	858	B21-3×	Idealized ^d	34.7 psu	5,000		✓
S20 ^e	286	S20-1×	B21-1×	34.7 psu	2,000	✓	
	858	S20-3×	B21-3×	34.7 psu	2,000	✓	
	858	S20-3×-sal	B21-3×	B21-3×	2,000	✓	✓
Preindustrial	286	PI	D06 ^f	D06 ^f	2,000	✓	

^aAMOC: Atlantic Meridional Overturning Circulation. ^bPMOC: Pacific Meridional Overturning Circulation. Here defined as cross-hemispheric advection of North Pacific Deep Water. ^cB21: The Burls et al. (2021) paleogeography configuration. ^dIdealized temperature profiles from Steinig et al. (2024) and Lunt et al. (2021). ^eS20: The Straume et al. (2020) paleogeography configuration. ^fPre-industrial (PI) initial temperature and salinity fields from Delworth et al. (2006).

- To separately address the effect of circulation, we ran an additional S20 simulation with different initial salinity conditions. These conditions give rise to a different ocean circulation for the same atmospheric CO₂ concentration and geography (Hutchinson et al., 2025a).

Unless specified otherwise, all these simulations were initialized with uniform salinity of 34.7 practical salinity units (psu) and idealized temperature fields constructed from previous GFDL CM2.1 simulations of the deep past (~55 Ma) (Hutchinson et al., 2025a). Other boundary conditions such as ice sheets and plant functional types were

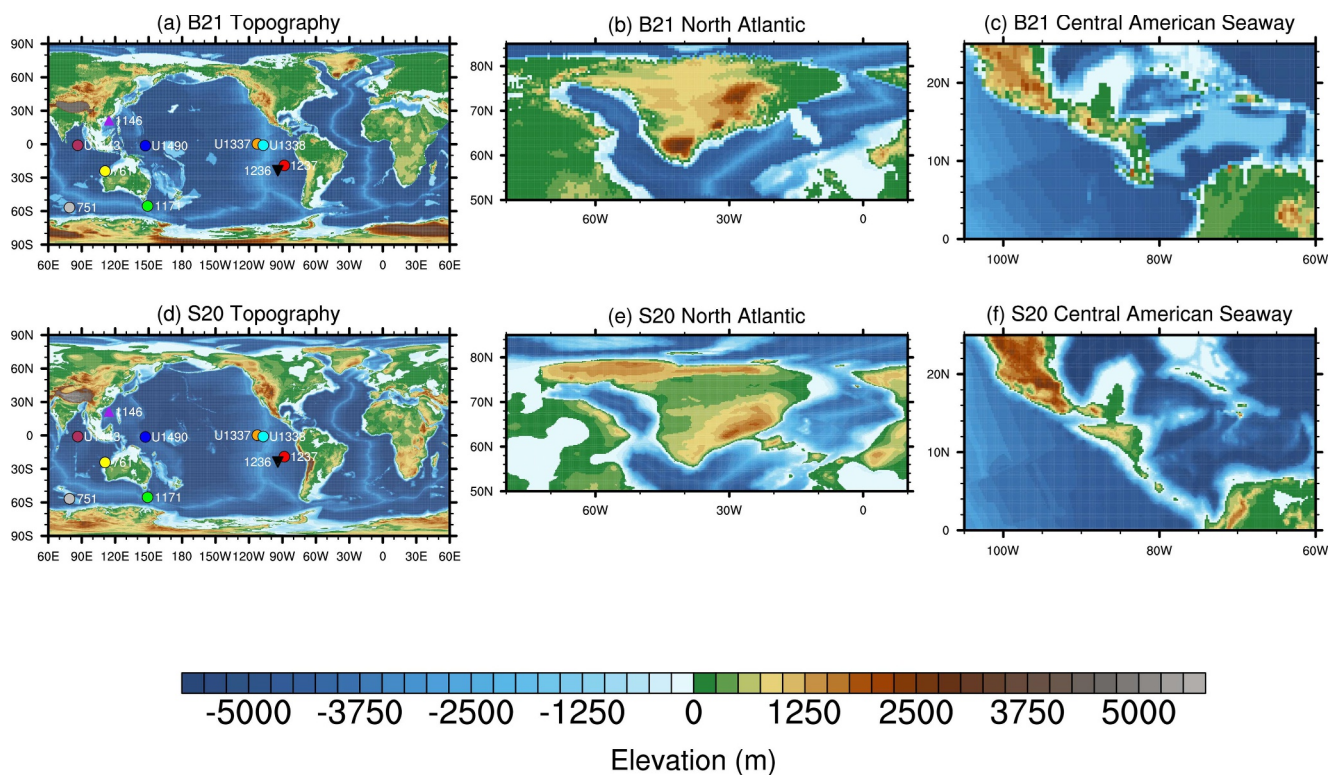


Figure 1. (a–c) B21 and (d–f) S20 15 Ma paleogeography, with plate-rotated Holbourn et al. (2024) drill site locations discussed in Section 3.2.1 overlaid, with triangle symbols representing sites identified by Holbourn et al. (2024) as having distinct watermass characteristics. (b, e) North Atlantic-Arctic gateway regions and (c, f) Central American Seaway (CAS). The (e) S20 paleobathymetry includes a more pronounced Greenland Scotland Ridge, a narrower gateway between Greenland and Europe, and an open North-West Passage between Greenland and Canada. The S20 paleogeography also possesses a shallower CAS (f) than the B21 configuration (c), with S20 depths ranging between 500 and 1,300 m in this region, while B21 depths are consistently greater than 1,500 m and exceed 5,000 m in some grid cells. Both paleogeographies are presented in their original resolutions. Text S1.1 in Supporting Information S1 describes and contrasts the two paleogeographic reconstructions.

specified by MioMIP1 (Burls et al., 2021), and we used constant preindustrial aerosol forcings, solar constant, and orbital parameters.

We conducted a preindustrial control simulation (available through the Deep-Time Model Intercomparison database; Steinig et al., 2024), with initial temperature and salinity taken from the preindustrial (year 1860) simulation of Delworth et al. (2006).

We note that our Miocene simulations can also be interpreted in a sequence of decreasing CO₂ as representing different episodes of the middle Miocene, namely the transition between the warmer MCO and the cooler MMCT. That is, we can interpret the 3× and 2×CO₂ simulations to represent the MCO and the 1×CO₂ to represent the MMCT, which informs us on middle Miocene climatic changes.

2.2. Ocean Transport Matrices

To investigate ocean oxygenation during the Miocene, we use a simple biogeochemistry model that is driven by transport matrices that represent the annual mean ocean circulation of our GFDL CM2.1 simulations. Using transport matrices is computationally advantageous and allows us to directly compute the steady-state biogeochemistry that is in equilibrium with the average physical ocean state of GFDL CM2.1.

The ocean circulation transport operator becomes a sparse matrix (the “transport matrix”) when 3D grid values are organized into column vectors. The tracer equation for concentration $\chi(\mathbf{r})$ at location $\mathbf{r}$ is given by:

$$\frac{\partial \chi(\mathbf{r})}{\partial t} + \mathcal{T}(\mathbf{r})\chi(\mathbf{r}) = S(\mathbf{r}), \quad (1)$$

where t is time, $\mathcal{T}(\mathbf{r}) \equiv \mathbf{u}(\mathbf{r}) \cdot \nabla - \nabla \cdot \mathbf{K}(\mathbf{r})\nabla$ the advective-eddy-diffusive transport operator (with velocity field $\mathbf{u}$ and diffusivity tensor $\mathbf{K}$), and $S(\mathbf{r})$ the net sources minus sinks of χ . The corresponding matrix equation for the column vector χ is

$$\frac{\partial \chi}{\partial t} + \mathbf{T}\chi = \mathbf{S}, \quad (2)$$

where $\mathbf{T}$ is the transport matrix and $\mathbf{S}$ is the column vector of sources minus sinks.

The transport matrices are built from GFDL CM2.1 outputs using the method of Chamberlain et al. (2019) and Pasquier et al. (2024). Horizontal advection is taken from monthly volume fluxes averaged over the last 100 years of our GFDL CM2.1 simulations. These volume fluxes include both resolved and parameterized eddy components (Gent & McWilliams, 1990). Vertical advection is then computed through mass conservation from the seafloor up to the surface, and thus captures implicit free surface variations.

Diffusion is added explicitly through tuned parameters following the method of Chamberlain et al. (2019). It includes constant background horizontal mixing (with a uniform diffusivity coefficient of 500 m² s^{−1}), background vertical mixing (4.5 × 10^{−5} m² s^{−1}), and stronger vertical mixing in the mixed-layer (0.1 m² s^{−1}). These coefficients were adjusted from those of Chamberlain et al. (2019) to improve the match between the ideal mean water age calculated offline with the transport matrix against the age calculated online with GFDL CM2.1 (see Figure S1 in Supporting Information S1).

2.3. PCO₂ Biogeochemistry Model

PCO₂ is an offline biogeochemical model developed by Pasquier et al. (2023). It includes 11 dissolved or particulate tracers that track the coupled cycles of phosphorus, carbon, and oxygen through the ocean (see Text S1.2 in Supporting Information S1 for a detailed description of the model). For each GFDL CM2.1 simulation, we compute the steady biogeochemical state using the same Newton solver as Pasquier et al. (2023).

That is, we stack the column vectors of the 11 model tracers into a single vector, set each tendency to zero, and compute the coupled steady state by solving the corresponding nonlinear problem.

The steady state that we solve for is idealized, in the sense that we impose boundary conditions for temperature, salinity, sea ice, and photosynthetically available radiation. Global inventories of phosphate and alkalinity are

prescribed by restoring their concentrations toward a global average based on observational pre-industrial values (this is necessary to avoid a singular Jacobian).

2.3.1. Matrix Biogeochemical Solution

Mathematically, we solve $\mathbf{F}(\mathbf{x}) = 0$ where $\mathbf{F}(\mathbf{x}) = \partial\mathbf{x}/\partial t$ is the vector of the biogeochemical tracer tendencies. Newton's method does this iteratively from an initial guess $\mathbf{x}_0$ by updating it via

$$\mathbf{x}_{n+1} = \mathbf{x}_n - \mathbb{J}(\mathbf{x}_n)^{-1} \mathbf{F}(\mathbf{x}_n), \quad (3)$$

where $\mathbb{J}(\mathbf{x}) = \partial\mathbf{F}(\mathbf{x})/\partial\mathbf{x}$ is the Jacobian matrix of $\mathbf{F}$. In practice, $\mathbb{J}$ is decomposed as

$$\mathbb{J}(\mathbf{x}) = \mathbb{T} + \mathbb{D}(\mathbf{x}) \quad (4)$$

where $\mathbb{T}$ is a block diagonal matrix of the transport matrices ($\mathbf{T}$) and $\mathbb{D}(\mathbf{x})$ arises from the local sources and sinks (a block matrix of diagonal matrices of the derivatives of the tracer sources and sinks, such as PO_4 uptake, remineralization, dissolution, and so on). In theory, providing $\mathbf{F}$ and $\mathbb{J}$ is sufficient for solving for the steady state. However, convergence is not always guaranteed in practice, and the nonlinear solver used in PCO2 was customized specifically for this biogeochemistry model.

2.3.2. Oxygen Tracer Equation

Oxygen concentrations (O_2) are governed by the following tracer equation:

$$\frac{\partial \text{O}_2}{\partial t} + \tau \text{O}_2 = r_{\text{O}_2:\text{C}} [U_{\text{C}} - (R_{\text{DOC}} + R_{\text{POC}}) \Theta(\text{O}_2 - \text{O}_2^{\text{lim}})] + J_{\text{O}_2}^{\text{atm}}, \quad (5)$$

where we have omitted the location $\mathbf{r}$ for simplicity.

The right-hand-side of Equation 5 models the O_2 sources and sinks. U_{C} corresponds to biological carbon uptake, which is then multiplied by the stoichiometric ratio of oxygen to carbon, $r_{\text{O}_2:\text{C}}$, which is allowed to vary spatially as per Galbraith and Martiny (2015). R_{DOC} represents the remineralization of dissolved organic carbon (DOC) and R_{POC} the respiration of particulate organic carbon (POC). Oxygen consumption ceases below the threshold concentration $\text{O}_2^{\text{lim}} = 5 \mu\text{mol kg}^{-1}$ through the Heaviside step function Θ , 1 for positive values and 0 for negative values. Wind- and temperature-dependent atmosphere-ocean exchange of oxygen is represented by $J_{\text{O}_2}^{\text{atm}}$, which follows Wanninkhof (2014).

2.3.3. Oxygen and Water Source Diagnostics

To disentangle the effects of solubility, circulation, and biology on oxygen concentrations, we partition oxygen concentration into a preformed component (O_2^{pre}) and True Oxygen Utilization (TOU). That is, we use

$$\text{O}_2 = \text{O}_2^{\text{pre}} - \text{TOU}, \quad (6)$$

where O_2^{pre} represents the oxygen concentration expected at each grid cell with no respiration in the ocean interior, and TOU represents the oxygen that has been cumulatively removed by respiration after leaving the euphotic zone (see, e.g., Ito et al., 2004).

Preformed O_2 can be thought of as the surface oxygen concentrations propagating conservatively in the ocean interior. Below the euphotic zone, the tracer equation for O_2^{pre} is thus simply

$$\left(\frac{\partial}{\partial t} + \tau\right) \text{O}_2^{\text{pre}} = 0, \quad (7)$$

subject to the boundary condition that $\text{O}_2^{\text{pre}} = \text{O}_2$ in the euphotic zone. We follow Pasquier et al. (2024) and compute O_2^{pre} directly as $\text{O}_2^{\text{pre}} = (\mathbf{T} + \mathbf{L})^{-1} \mathbf{L} \text{O}_2$, where $\mathbf{L}$ is a diagonal matrix with $1/\tau$ entries for cells in the

euphotic zone (we use $\mathbf{L}$ with timescale $\tau = 1$ s as a convenient way to reproduce the boundary conditions for both O_2^{pre} and TOU). A given cell's O_2^{pre} is therefore determined by a combination of solubility and ventilation, reflecting the saturated oxygen concentration possible at the cell in concert with the saturated oxygen concentration at the surface point of origin. This differs from, for example, saturated oxygen concentration (O_2^{sat}), which reflects only the maximum solubility-enabled oxygen concentration at a given grid cell.

Conversely, TOU can be thought of as the accumulation of respiration, that is, Oxygen Utilization Rates (OUR), over time. Below the euphotic zone, TOU is thus simply governed by

$$\left(\frac{\partial}{\partial t} + \tau\right) \text{TOU} = \text{OUR}, \quad (8)$$

subject to the boundary condition that $\text{TOU} = 0$ in the euphotic zone. Similarly to O_2^{pre} , we follow Pasquier et al. (2024) and compute the vector of TOU directly as $\text{TOU} = (\mathbf{T} + \mathbf{L})^{-1} \text{OUR}$. TOU therefore is a combination of the rate of oxygen utilization through biological processes and water mass residence time (Holzer, 2022).

The changes in O_2^{pre} and TOU between two simulations can also be partitioned following the framework of Pasquier et al. (2024).

We decompose changes in the domain-averaged inventory of O_2^{pre} into contributions from solubility, circulation, and their nonlinear interaction:

$$\Delta O_2^{\text{pre}} = \Delta(O_2 \mathcal{V}_{\Omega}^{\text{l}}) = \underbrace{\mathcal{V}_{\Omega}^{\text{l}} \Delta O_2^{\text{sat}}}_{\text{solubility}} + \underbrace{O_2^{\text{sat}} \Delta \mathcal{V}_{\Omega}^{\text{l}} + \Delta(O_2^{\text{dis}} \mathcal{V}_{\Omega}^{\text{l}})}_{\text{circulation}} + \underbrace{\Delta O_2^{\text{sat}} \Delta \mathcal{V}_{\Omega}^{\text{l}}}_{\text{cross term}}. \quad (9)$$

where $\mathcal{V}_{\Omega}^{\text{l}}$ represents the fraction of the volume of a given domain Ω (a subvolume of the ocean) ventilated per unit area at the base of the euphotic zone. O_2^{sat} is the oxygen concentration corresponding to the maximum dissolved oxygen permitted by local solubility. O_2^{dis} is the disequilibrium component of O_2^{pre} such that $O_2^{\text{pre}} = O_2^{\text{sat}} + O_2^{\text{dis}}$, reflecting the variation of air-sea O_2 disequilibrium with location. A “ Δ ” preceding a term indicates difference between two simulations, and the absence of a Δ signifies that the value is taken from the reference state (here, the $1\times\text{CO}_2$).

Similarly, changes in TOU are decomposed into contributions from respiration, circulation, and their nonlinear interaction:

$$\Delta \text{TOU} = \Delta(\text{OUR} \times \Gamma_{\Omega}^{\text{r}}) = \underbrace{\Gamma_{\Omega}^{\text{r}} \Delta \text{OUR}}_{\text{respiration}} + \underbrace{\text{OUR} \Delta \Gamma_{\Omega}^{\text{r}}}_{\text{circulation}} + \underbrace{\Delta \text{OUR} \Delta \Gamma_{\Omega}^{\text{r}}}_{\text{cross term}}. \quad (10)$$

where $\Gamma_{\Omega}^{\text{r}}$ denotes the cumulative exposure time of a parcel of water to upstream respiration prior to entering the domain Ω .

These decompositions allow us to attribute oxygen changes between high- and low- CO_2 simulations (evaluated relative to the $1\times\text{CO}_2$ reference state) to linear contributions from solubility, ventilation pathways (O_2^{pre}), respiration, and water mass residence times (TOU), as well as to second-order interaction terms between each pair of processes.

Finally, the water mass composition at any location can be accurately computed using our transport matrices, following, for example, Holzer et al. (2020). The fraction of water that was last in contact with the surface region Ω is given by

$$f_{\Omega}^{\text{l}} = (\mathbf{T} + \mathbf{L})^{-1} \mathbf{L} \mathbf{\Omega}, \quad (11)$$

where the vector $\mathbf{\Omega}$ is equal to 1 in region Ω and zero everywhere else.

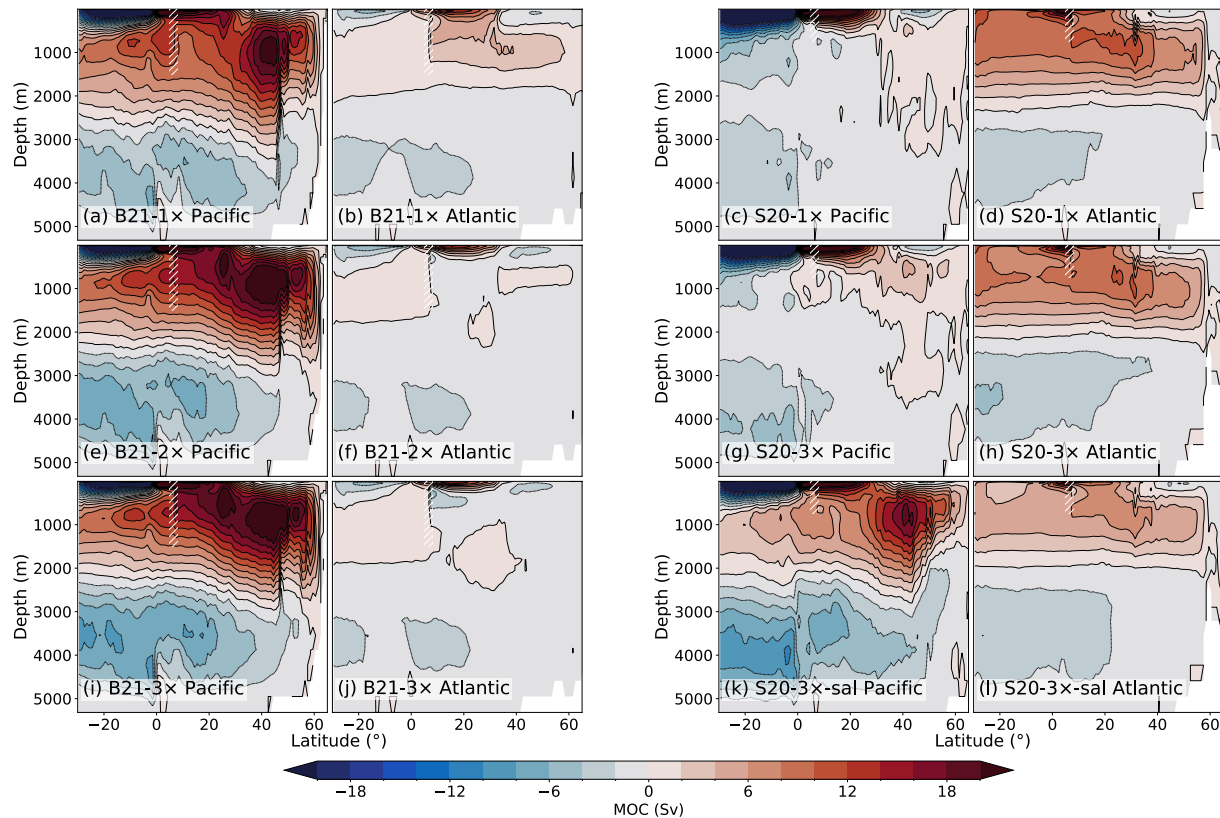


Figure 2. Atlantic and Pacific overturning circulations in Sverdrups (Sv) for all simulations shown between 30°S and 65°N. B21 paleogeography utilized in (a, b) B21-1x, (e, f) B21-2x, (i, j) B21-3x. S20 paleogeography in (c, d) S20-1x, (g, h) S20-3x, and (k, l) S20-3x-sal. Global MOC streamfunctions and differences are shown in Hutchinson et al. (2025a). Hatching marks zonally unbounded latitudes where the overturning streamfunction is undefined.

3. Results

3.1. Large-Scale Ocean Circulation

Our Miocene simulations exhibit distinct modes of meridional overturning circulation (MOC) depending on paleogeography and initial conditions (Table 1 and Figure 2). All B21 configurations exhibit an active cross-hemispheric Pacific Meridional Overturning Circulation (PMOC), but only the low- CO_2 simulation (B21-1x) sustains an Atlantic Meridional Overturning Circulation (AMOC). This can be thought of as the AMOC collapsing with increasing CO_2 . Conversely, all the S20 simulations present an active AMOC, but only S20-3x-sal develops an active cross-hemispheric PMOC.

Paleogeographic differences explain why the S20 simulations all exhibit a strong AMOC while it collapses in the B21 simulations at high CO_2 . In all configurations, higher CO_2 concentrations and warmer temperatures result in stronger precipitation minus evaporation (P-E), and stronger river discharge at high latitudes, freshening the surface water and increasing stratification in the Arctic Ocean (Hutchinson et al., 2025a). The large gateway between the Arctic and the North Atlantic in the B21 reconstruction (Figure 1b) allows this light fresh water to flow into the North Atlantic and prevents deep water formation in the 2x and 3x CO_2 simulations. This gateway is substantially smaller in the S20 configuration (Figure 1e). Hence, the North Atlantic surface remains saline enough for deep water to form in all S20 simulations, even at 3x CO_2 .

In all configurations, increased CO_2 weakens the AMOC but strengthens the PMOC (Figure 2). Maximum PMOC transport intensifies by 2.8 Sverdrups (Sv) between B21-1x and B21-3x, while the already weak AMOC in B21-1x shuts down for higher CO_2 concentrations. When integrated with S20 topography, the maximum AMOC strength weakens by a trivial 0.2 Sv between S20-1x and S20-3x and by 3.4 Sv between S20-1x and S20-3x-sal.

Two stable circulation equilibria exist in the S20 geographic configuration with $3\times\text{CO}_2$. One equilibrium has an active cross-hemispheric PMOC (S20-3 $\times$ -sal; Figure 2k), the other a much reduced Pacific deepwater formation (S20-3 $\times$; Figure 2g). These two equilibria were reached from different initial conditions. Starting from a “salty” North Pacific produces and sustains a PMOC, while starting from a uniform salinity does not produce a PMOC. S20-3 $\times$ sees a slight increase in the strength of Pacific circulation when compared to S20-1 $\times$, increasing by 2.4 Sv, while the S20-3 $\times$ -sal simulation displays Pacific circulation 15.9 Sv stronger than S20-1 $\times$. The strength of the AMOC is weaker when PMOC is active (Hutchinson et al., 2025a).

The high- CO_2 simulations present more vigorous Antarctic Bottom Water (AABW) formation than their low- CO_2 counterparts (Hutchinson et al., 2025a). In the B21 simulations, higher CO_2 levels lead to significantly increased AABW formation in the Pacific sector, and decreased AABW formation in the Atlantic sector (Figure 2; Hutchinson et al., 2025a). The AABW cell is stronger in the Pacific basin than the Atlantic in each B21 simulation. Similarly, S20-3 $\times$ and especially S20-3 $\times$ -sal exhibit stronger Pacific AABW formation than S20-1 $\times$ (Figure 2). In contrast to the B21 paleogeography, the strength of Atlantic AABW formation also increases slightly between the 1 $\times$ and $3\times\text{CO}_2$ S20 simulations. In both sets of simulations, there is a southward shift in AABW convection sites at higher CO_2 , due to sea ice retreat in these regions (Figure S2 in Supporting Information S1).

In both paleogeographies, Antarctic Intermediate Water (AAIW) formation decreases at higher CO_2 (Figure S2 in Supporting Information S1), and AAIW formation sites tend to shift poleward. However, we observe localized increases in AAIW formation in the proto-Tasman Sea in each 3 $\times$ simulation, which is particularly pronounced in B21-3 $\times$ and S20-3 $\times$.

3.2. Comparison to Proxies

There is some evidence for an active PMOC in the early Cenozoic (Ferreira et al., 2018; McKinley et al., 2019; Thomas et al., 2014). After the middle Miocene, however, PMOC becomes less prominent until the Pliocene, when there is evidence for a resurgence of cross-hemispheric PMOC until ~ 2.73 Ma (Burls et al., 2017).

It is uncertain when exactly the PMOC might have ceased (Ferreira et al., 2018; Naik et al., 2025). However, Miocene $\delta^{13}\text{C}$, ϵ_{Nd} , and biomarker proxies recorded evidence for a major shift in global ocean circulation between the MCO and MMCT (Holbourn et al., 2022, 2024; Kender et al., 2014, 2018; Lübbers et al., 2019; Super et al., 2020). For example, ϵ_{Nd} signatures from an abyssal drill site in the Philippine Sea reveal that during the MCO, the site was influenced by radiogenic North Pacific Deep Water (NPDW), which was later replaced by unradiogenic Circumpolar Deep Water (CDW), signaling a transition from a PMOC-like circulation regime to a more modern-like Pacific circulation during the MMCT (Kender et al., 2018). Our model broadly captures these dynamics. The higher CO_2 simulations, which we treat as approximate analogs to the MCO, involve a more vigorous Pacific overturning circulation, with stronger ventilation in the Pacific than the colder (MMCT analog) simulations for both continental configurations.

Super et al. (2020) used sediment biomarkers to assess Atlantic temperatures during the Miocene, finding a weakening of the meridional Atlantic sea surface temperature (SST) gradient across the MMCT. The authors attribute this change to a strengthening of the AMOC or North Atlantic Current. Our simulations qualitatively reproduce the association between MMCT cooling and a stronger AMOC. Overturning is weaker at higher CO_2 in both suites, but the response differs in degree: the AMOC collapses in B21-2 $\times$ and B21-3 $\times$ relative to B21-1 $\times$, whereas the S20 simulations retain an active AMOC throughout, with S20-3 $\times$ -sal's AMOC being substantially weaker than S20-1 $\times$ (3.4 Sv), and the AMOC in S20-3 $\times$ being marginally weaker than in S20-1 $\times$ (0.2 Sv).

The paleo-records presented by Super et al. (2020) include TEX_{86} and U_{37}^{K} data. The TEX_{86} reconstruction indicates that from the MCO to the MMCT, SST decreases by $\sim 5.5^\circ\text{C}$ at mid-latitude Deep Sea Drilling Project (DSDP) Site 608 but only by $\sim 2.1^\circ\text{C}$ at subpolar Ocean Drilling Program (ODP) Site 982, corresponding to a gradient change of -3.4°C (Site 608–Site 982). Modeled SST changes from B21-3 $\times$ to B21-1 $\times$ are -3.66°C at the location of Site 608 and -0.64°C at the location of Site 982, giving a gradient change of -3.02°C between MCO-analog and MMCT-analog, within 0.4°C of the reconstructed value. Surprisingly, Site 982 warms by 0.86°C across the B21-2 $\times$ –B21-1 $\times$ transition, indicating that the collapse of the AMOC at $2\times\text{CO}_2$ leads to a regional cooling despite the higher CO_2 forcing (Figure S3 in Supporting Information S1).

The S20 simulations show a more uniform meridional response due to their active AMOC: for the same transition ($1\times$ minus $3\times$ in each case), SST changes by -3.41°C at the location of Site 608 and by -3.82°C at the location of Site 982 for S20- $3\times$, and by -3.24°C and -2.46°C for S20- $3\times$ -sal (Figure S3 in Supporting Information S1). These changes correspond to meridional temperature gradient changes between the two sites of $+0.41^{\circ}\text{C}$ and -0.78°C . The B21 ensemble agrees better with the reconstructed temperature gradient changes. S20- $3\times$ -sal also captures a small negative change (the direction of the proxy reconstruction), while in S20- $3\times$ the meridional temperature gradient increases. Our B21 simulations therefore corroborate the findings of Super et al. (2020), namely a weaker North Atlantic circulation in the MCO-analogs.

While our analysis above is based on anomalies, we note that all our simulations substantially underestimate absolute TEX_{86} - and $U_{37}^{K'}$ -reconstructed SSTs. This is consistent with prior findings that climate models, including GFDL CM2.1, generally underestimate high-latitude SSTs when simulating the Miocene and other high- CO_2 climates at proxy-constrained CO_2 levels (Burls et al., 2021; Huber & Caballero, 2011; Hutchinson et al., 2025a; Lunt et al., 2021).

3.2.1. Water Mass Characteristics in the Indo-Pacific Oceans

Holbourn et al. (2024) showed that sediment cores in the Pacific and Indian Oceans exhibit similar $\delta^{13}\text{C}$ values during the MCO, which are indicative of a shared southern-sourced water mass. The authors concluded that during the MCO, ODP and International Ocean Discovery Program (IODP) sediment core sites in the Western Tropical Pacific (Site U1490), Southern Ocean (Sites 1171, 751), Indian Ocean (761, U1443), and Eastern Pacific (1236, 1237, U1338, and U1337) were bathed in the same cold, fresh, and southern-sourced water mass. In contrast, a northern subtropical site in the South China Sea (1146) presents lower $\delta^{18}\text{O}$ and $\delta^{13}\text{C}$ signatures. For site depths and paleocoordinates see Table S1 in Supporting Information S1 and Figure 1.

We analyze simulated potential temperature (T) and salinity (S) characteristics at the paleo-locations and depths of these sites (Figure 3). The paleo-locations were determined by rotating the drill sites' modern coordinates back to 15 Ma using the plate rotation model of Müller et al. (2008), which is the same model that was used to construct the B21 topography (locations are shown in Figure 1). All results from locations of drill sites that were described in Holbourn et al. (2024) as sharing similar water characteristics at the MCO are represented as circles in Figure 3, with the exception of Site 1236, which is represented as a downward triangle. We also plotted the T-S characteristics of grid boxes that are in regions of southern- ("southern ocean cloud," green) and northern-sourced ("north pacific cloud," pink) water masses. We find that for each simulation at least some of the locations represented by circles are bathed in, or similar in characteristic to, AABW (near the green clouds in Figure 3).

Several simulations can also plausibly capture the distinct water-mass characteristics inferred from the $\delta^{18}\text{O}$ and $\delta^{13}\text{C}$ signatures of Site 1146 in the South China Sea (represented in Figure 3 by an upward triangle). Site 1146 is on a temperature or salinity extreme and closer to northern-sourced water mass characteristics (pink clouds in Figure 3) in all S20 simulations and in B21- $2\times$ and B21- $3\times$. In simulations S20- $3\times$ and S20- $3\times$ -sal this location is particularly fresh, whereas in B21- $2\times$ and B21- $3\times$ it is the most saline, but also one of the warmest sites. In three of the warm simulations (B21- $2\times$, B21- $3\times$, and S20- $3\times$ -sal) Site U1490 shares these northern-sourced characteristics with site 1146, contrary to Holbourn et al. (2024)'s hypothesis of Site U1490 being bathed in a separate, southern-sourced water mass.

Holbourn et al. (2024) also showed that the water mass bathing an intermediate-depth site in the Eastern Tropical Pacific (1236) was similar to the other sites during the MCO, but diverged during the MMCT. In all of our simulations, simulated potential temperatures and salinities at the location of Site 1236 (represented by a downward triangle in Figure 3) are somewhat distinct from the others. The smallest differences are found in B21- $2\times$ and B21- $3\times$. T-S characteristics at Site 1236 diverge from the other locations in B21- $1\times$ relative to B21- $2\times$ and B21- $3\times$, which is consistent with the findings of Holbourn et al. (2024).

We also analyze steady-state water source fraction at these locations. Figure 4 shows the percentage of the water bathing each drill site that is made up of North Pacific- and Southern Ocean-sourced water. These two water masses are the dominant water masses at each site, with the remainder largely made up of recirculated modal and intermediate waters. This analysis corroborates the results of Holbourn et al. (2024) and shows that Site 1146 (upward triangle) is predominantly bathed in northern-sourced waters in all of our B21 simulations and in

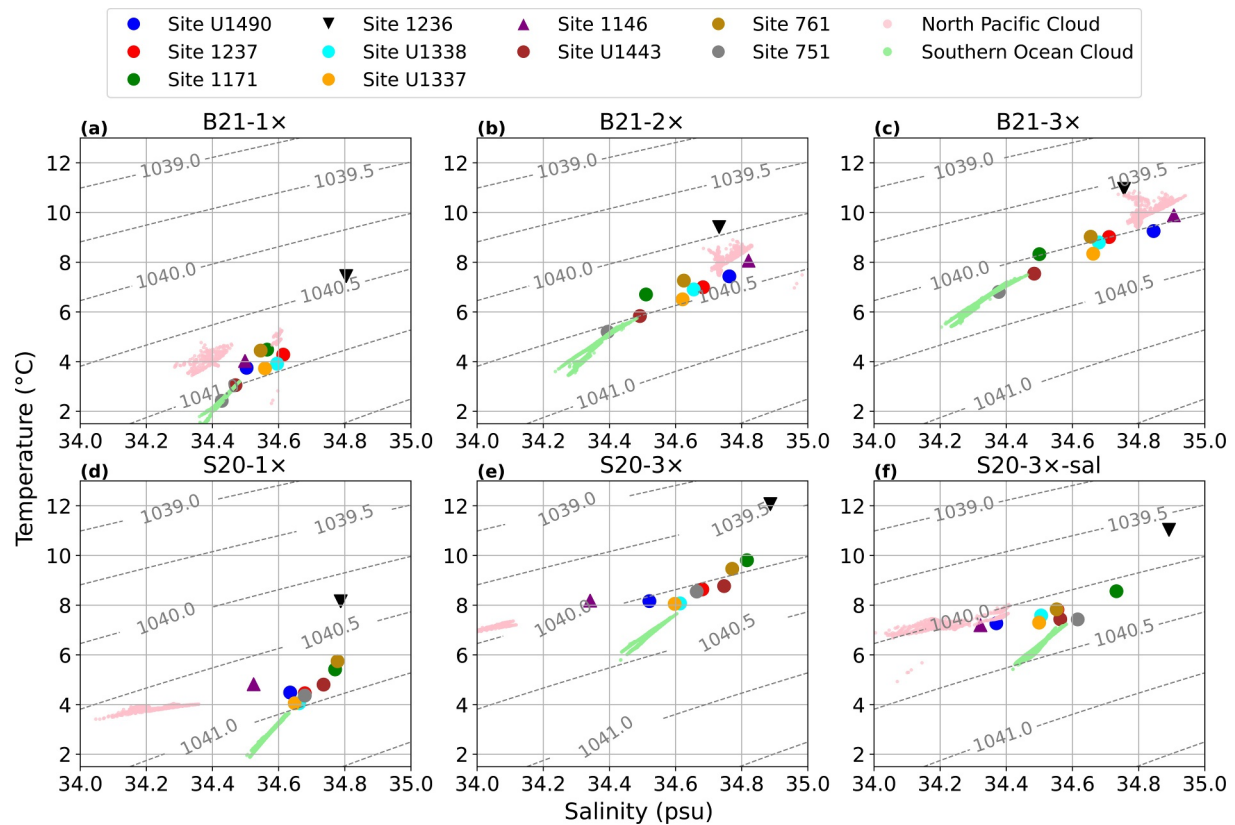


Figure 3. Temperature and salinity characteristics for (a) B21-1x, (b) B21-2x, (c) B21-3x, (d) S20-1x, (e) S20-3x, and (f) S20-3x-sal simulations at the approximate 15 Ma drill site locations described in Holbourn et al. (2024) and shown in Figure 1, with 3,000 m isopycnals overlaid. Clouds of simulated North Pacific (pink, latitude north of 55°N, 1,500–2,000 m depth) and Southern Ocean (green, latitude south of 60°S, 3,500–4,000 m depth) potential temperatures and salinities are included for reference for each simulation.

S20-3x-sal. Furthermore, most of the other drill site locations are bathed predominantly in southern-sourced waters in our S20 simulations, in agreement with Holbourn et al. (2024). S20-1x exhibits a particularly close clustering of the sites, with almost no northern-sourced water masses, but it is not an MCO-analog simulation. At Site U1490, our B21 runs and S20-3x-sal simulate predominantly northern-sourced water masses, which explains its proximity in terms of T-S characteristics to Site 1146 (Figure 3) but is in disagreement with Holbourn et al. (2024).

Holbourn et al. (2024) inferred that Indo-Pacific sites were ventilated by a common southern-sourced water mass during the MCO, with Site 1146 in the proto-South China Sea recording a distinct northern-sourced signal. Our simulations broadly reproduce this structure when evaluated jointly in T-S space (Figure 3) and by water source fractions (Figure 4). Across all experiments, Indo-Pacific sites cluster with Southern Ocean-sourced waters, while Site 1146 tends toward North Pacific characteristics. This northern signal is strongest in the B21 suite and S20-3x-sal and weaker in S20-3x, a pattern consistent across both diagnostics. However, B21-2x, B21-3x, and S20-3x-sal also assign Site U1490 to northern sources, contrary to proxy interpretations. By contrast, S20-3x preserves the proxy-inferred separation, maintaining a northern signal at Site 1146 while keeping Site U1490 aligned with southern-sourced Indo-Pacific sites.

3.2.2. Carbonate Ion Concentrations

Analysis of carbonate ion (CO_3^{2-}) concentrations (Holbourn et al., 2022; Kender et al., 2014; Lübbers et al., 2019) indicates that CO_3^{2-} levels were reduced during the warm MCO and subsequently increased during the cooler MMCT. In particular, Kender et al. (2014) posit an increase of $40 \mu\text{mol kg}^{-1}$ in intermediate to deep water masses based on foraminiferal B/Ca ratios from numerous Atlantic, Pacific, Indian, and Southern Ocean sites, reflecting a pronounced shift in ocean carbonate chemistry. This change is mirrored in our model simulations, which show

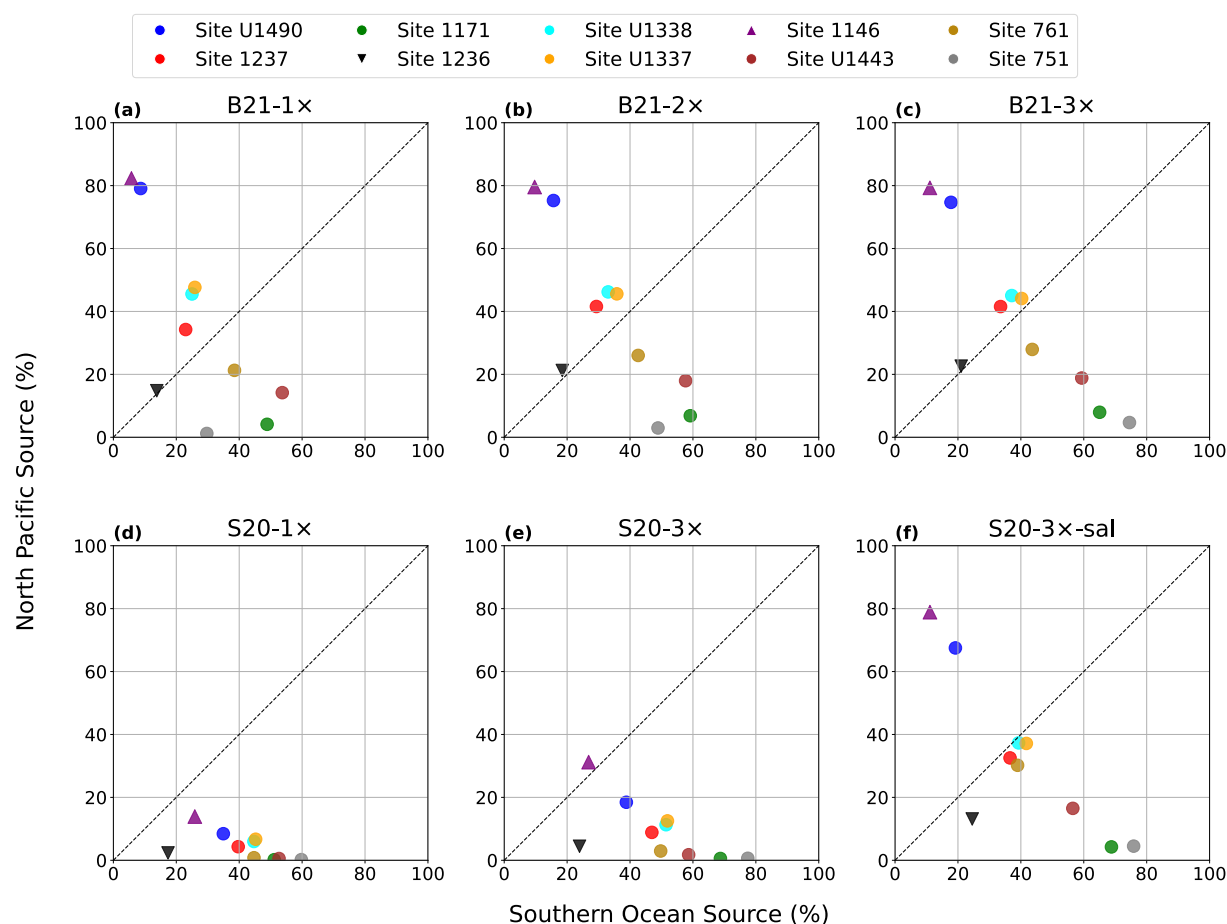


Figure 4. Percentages of Southern Ocean- (x -axis) and North Pacific-sourced (y -axis) water at the approximate drill site locations described in Holbourn et al. (2024) for (a) B21-1 $\times$, (b) B21-2 $\times$, (c) B21-3 $\times$, (d) S20-1 $\times$, (e) S20-3 $\times$, and (f) S20-3 $\times$ -sal.

significantly lower CO_3^{2-} concentrations throughout intermediate and deep waters in B21-3 $\times$ compared to B21-1 $\times$ (global mean decrease of $\sim 63 \mu\text{mol kg}^{-1}$, Figure S4 in Supporting Information S1). A similar pattern emerges in the S20 simulations, where S20-3 $\times$ and S20-3 $\times$ -sal exhibit substantially less CO_3^{2-} in intermediate and deep waters (global mean decrease of ~ 62 and $\sim 65 \mu\text{mol kg}^{-1}$, Figure S4 in Supporting Information S1). This was expected because our simulations do not include the effect of enhanced weathering or marine sediment dissolution, which might offset the CO_3^{2-} changes associated with changes in atmospheric pCO_2 .

Kender et al. (2014) attribute the increase in dissolved CO_3^{2-} to changes in ocean alkalinity, rather than to shifts in ocean circulation or ecosystem dynamics. Our simulations are consistent with this hypothesis, as regional changes in CO_3^{2-} due to circulation changes are much smaller in our simulations than the mean change due to atmospheric CO_2 concentrations (Figure S4 in Supporting Information S1).

Sosdian et al. (2020) inferred elevated surface dissolved inorganic carbon (DIC) during the MCO to explain surface-proxy trends in B/Ca, $\delta^{11}\text{B}$, $\delta^{13}\text{C}$, and Mg/Ca. Consistent with these observations, our results show higher DIC concentrations in S20-3 $\times$, S20-3 $\times$ -sal, and B21-3 $\times$ relative to their low- CO_2 counterparts (not shown), mainly due to the higher imposed CO_2 boundary conditions.

3.2.3. Summary of Comparison to Proxies

Both the B21 and S20 simulations capture large-scale features of Miocene circulation, including enhanced North Atlantic overturning during the MMCT (Super et al., 2020), and stronger, deeper North Pacific overturning during

Table 2

Mean Dissolved O₂ Concentrations (μmol kg⁻¹)

Simulation	Global	Atlantic	Pacific	Southern Ocean ^a	Eastern tropical Pacific ^b
B21-1×	191	188	188	279	136
B21-2×	180	149	192	263	156
B21-3×	177	145	189	253	159
S20-1×	175	200	158	280	125
S20-3×	171	184	162	258	133
S20-3×	183	180	183	258	149
PI	172	219	141	285	89

^aSouth of 60°S. ^bEast of 150°W, within 30°S–30°N.

the MCO (Kender et al., 2018). Similarly, both sets of simulations capture the biogeochemical dynamics described by Kender et al. (2014), Lübbers et al. (2019), and Holbourn et al. (2022).

When comparing our simulations with Holbourn et al. (2024), B21-2×

Further evaluation of our simulations against proxy evidence is discussed in Sections 3.3.4 and 3.3.7, where we analyze modeled changes in dissolved oxygen in the Pacific Ocean.

3.3. Dissolved Oxygen Concentrations

3.3.1. Global Oxygen Response

Across all our simulations, global mean oxygen concentrations decline with increasing atmospheric CO₂, with the exception of S20-3×

To investigate the drivers of these differences in oxygen, we examine changes in O₂^{pre} and TOU for different ocean basins in Sections 3.3.2, 3.3.4, and 3.3.6. Changes in O₂^{pre} capture the combined effect of changes in solubility and ventilation pathways, while changes in TOU capture the effect of changes in respiration, modulated by changes in circulation via water mass residence times (see Methods Section 2.3.3).

When linking changes in TOU to changes in respiration rates, it is important to keep in mind that these changes are not necessarily colocated and/or correlated. TOU accumulates over the entire history of a water parcel since it left the surface. Consequently, respiration rates everywhere upstream of the flow, particularly close to the surface, can influence TOU anywhere, even near the seafloor. In addition, the circulation plays a crucial role through the upstream exposure time, which is the time over which respiration can act. This upstream exposure time can blur the link between TOU and OUR, and we emphasize that it is not the same as the age tracer (Holzer, 2022). Furthermore, nonlinear effects from colocated changes in upstream exposure time and OUR can also be important for large perturbations as conducted here. Such nonlinear effects are difficult to anticipate and to disentangle from first-order effects (Pasquier et al., 2024).

Preformed O₂ is often thought to be controlled by surface temperature and solubility only, but changes in O₂^{pre} can be more complicated to interpret. This is because the circulation can play a dominant role, as O₂^{pre} is also controlled by the pathways that connect the surface to the deep ocean. If ventilation patterns change in such a way

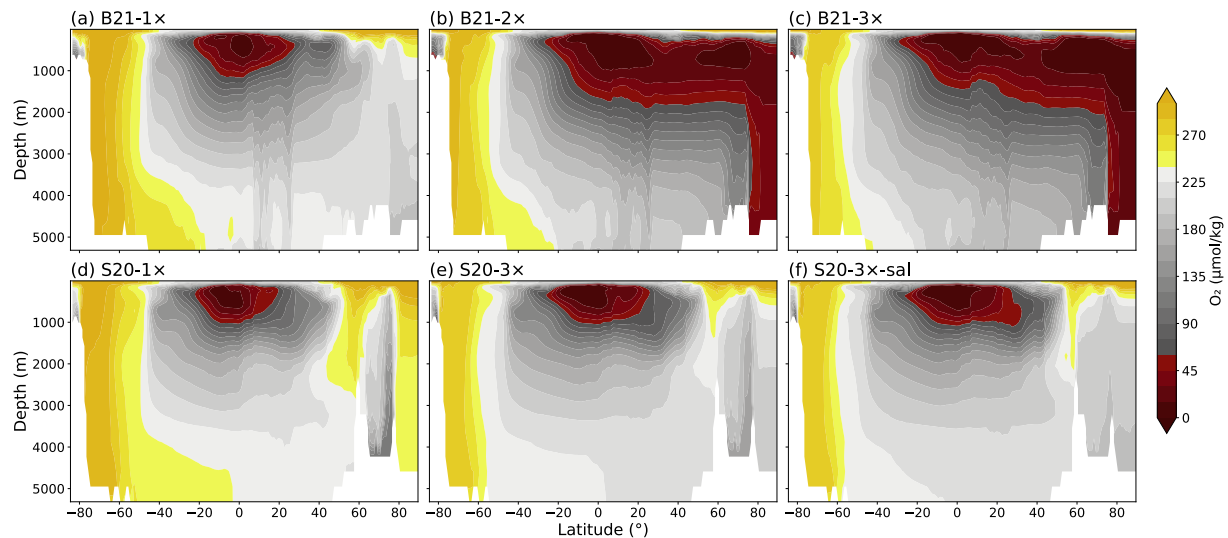


Figure 5. Atlantic and Arctic Ocean zonal average dissolved oxygen concentrations in $\mu\text{mol kg}^{-1}$ for (a–c) B21 and (d–f) S20 simulations.

that a given volume of interest Ω becomes ventilated by colder waters, then its O_2^{pre} will increase, even if temperatures are otherwise unchanged. Interpretation is further complicated by incomplete saturation at the surface, which can be caused by any combination of sea ice cover, short surface residence time, or upwelling of oxygen-poor waters. Both these effects are captured by the circulation component of $\Delta\text{O}_2^{\text{pre}}$ in Equation 9.

3.3.2. Atlantic and Arctic Ocean Oxygenation

In the Atlantic, oxygen concentrations vary with both paleogeography and CO_2 forcing. Consistent with the AMOC patterns and strengths shown in Figure 2, we find that S20-1x produces the highest basin-mean oxygen concentration of the Miocene simulations, while B21-3x exhibits the lowest (values reported in Table 2). Hypoxic volumes ($\text{O}_2 \leq 60 \mu\text{mol kg}^{-1}$, shown in red in Figure 5) expand with increasing CO_2 in all cases, but this effect is most pronounced when the AMOC is shut down in B21-2x and B21-3x. In contrast, the Atlantic remains well oxygenated in the S20 simulations, owing to the persisting AMOC.

In all geographic configurations, preformed O_2 generally decreases with increasing CO_2 (Figures 6b, 6f, and 6j), owing to warming and decrease in oxygen solubility. However, the decrease in O_2^{pre} is not uniform, and varies across configurations. For the B21 configuration, we find a larger O_2^{pre} decrease in Atlantic AABW, and in the regions bathed in North Atlantic Deep Water (NADW) in B21-1x, and a smaller decrease in intermediate and mode waters. For the S20 configuration, the O_2^{pre} decrease is present in Atlantic AABW but subdued in NADW. The differences in North Atlantic $\Delta\text{O}_2^{\text{pre}}$ between S20 and B21 can be explained by the AMOC/NADW formation persisting in the high- CO_2 S20 simulations but shutting down in B21-3x.

For B21-3x, the subdued decrease in O_2^{pre} at intermediate depths around 35°N is caused by the shutdown of warm and salty Tethys Intermediate Water (TIW) formation. TIW forms in B21-1x during winter and spring months (January–April, not shown), and intrudes into the Atlantic through the proto-Strait of Gibraltar, resulting in a relatively warm intermediate-depth North Atlantic (Figure S5 in Supporting Information S1). However, in B21-3x, atmospheric warming leads to a more stratified Tethys Ocean which prevents the formation of TIW. Without TIW, the northern subtropical Atlantic cools above 1,500 m, which modulates the global decrease in solubility. Although some Tethys convection occurs in S20-1x, it is less vigorous than in B21-1x and the difference between the 1x and 3x CO_2 simulations is not as pronounced.

Across all high- CO_2 simulations, upper-ocean TOU increases due to shoaled remineralization depth (Figure 7, Figure S6 in Supporting Information S1) and reduced upper-ocean ventilation (Figure S6 in Supporting Information S1). However, also consistent across these simulations, we find a decrease in TOU in the southern tropical latitudes (0 – 20°S) above 1,000 m (Figure 7). S20-3x and S20-3x-sal also show a decrease in TOU below approximately 2,000 m basin-wide (Figures 6g and 6k; note that the figure shows $-\Delta\text{TOU}$). This deep ocean TOU

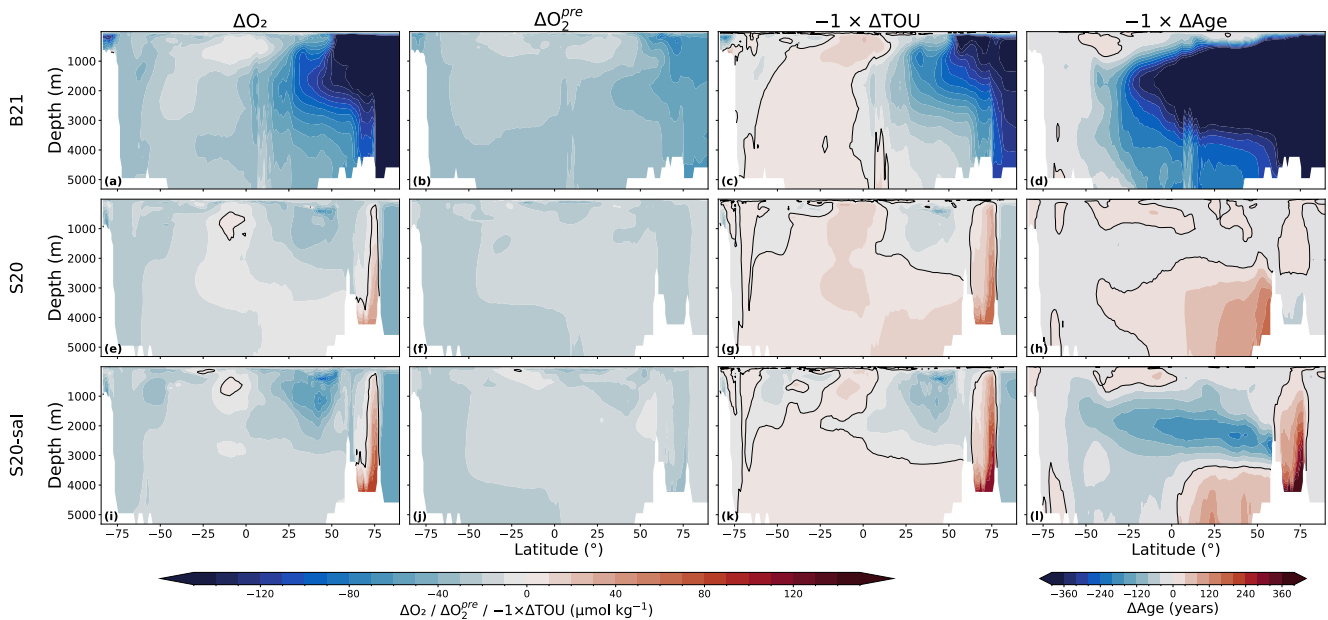


Figure 6. Difference in Atlantic and Arctic Ocean (a, e, i) zonal mean oxygen concentrations, (b, f, j) O_2^{pre} , (c, g, k) $-1 \times \Delta TOU$, in $\mu\text{mol kg}^{-1}$, and (d, h, l) $-1 \times \Delta \text{water age}$ in years for (a–d) B21-3X–B21-1X, (e–h) S20-3X–S20-1X, and (i–l) S20-3X–sal–S20-1X. We multiply TOU and water age by -1 to align the direction of change with their impact on oxygen concentration (see Section 2). The zero contour is plotted in black. B21-3X–B21-2X is shown in Figure S7 in Supporting Information S1.

decrease is also present in B21-3X, but limited to the southern hemisphere, due to compensating TOU increase in the northern hemisphere caused by AMOC shutdown (Figure 6c).

Decreases in TOU can be caused by decreased upstream OUR or by faster circulation reducing the time that a water mass was exposed to respiration (Pasquier et al., 2024). This deep TOU decrease reflects reduced interior basin remineralization under higher CO_2 in each simulation (Figure 7). Warming causes an upward shift in remineralization depth, enhancing surface and shallow remineralization while reducing it below 300 m.

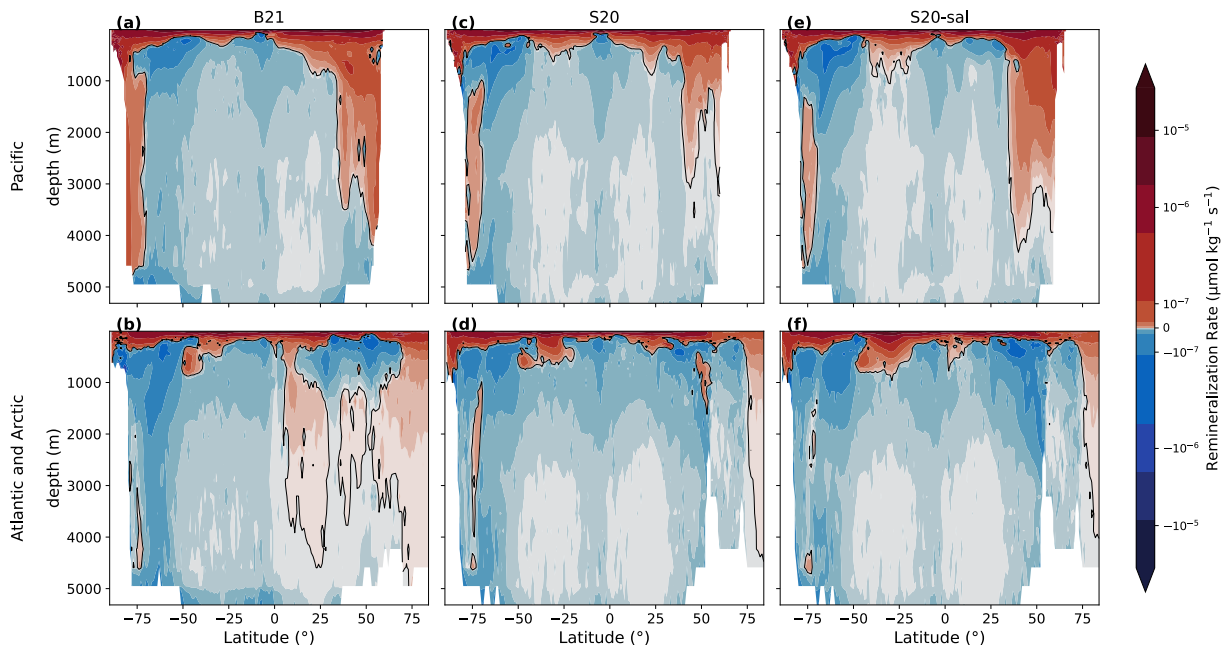


Figure 7. Zonal average remineralization rate anomalies in $\mu\text{mol kg}^{-1} \text{ s}^{-1}$ for (a, c, e) Pacific and (b, d, f) combined Atlantic and Arctic oceans. (a, b) B21-3X–B21-1X, (c, d) S20-3X–S20-1X, and (e, f) S20-3X–sal–S20-1X. The zero contour is plotted in black. Note pseudo-logarithmic color-scale.

Partitioning deep Atlantic TOU change into its biological and circulation-driven components, we see that each $3\times\text{CO}_2$ simulation exhibits decreased biological consumption of oxygen below 1,000 m. However, this remineralization-driven TOU decrease is overcome by circulation-driven TOU increase, resulting in a net TOU increase in B21-3 $\times$ and S20-3 $\times$ -sal (Figures S6c, S6f, and S6i in Supporting Information S1).

In contrast, Atlantic oxygen declines in the S20 simulations are primarily driven by O_2^{pre} loss, itself driven by solubility decrease (Figure 6, Figures S6d–S6i in Supporting Information S1). Nonetheless, TOU also plays a role: intermediate-depth North Atlantic TOU increases as the AMOC weakens (Figure 2), with the largest effect in S20-3 $\times$ -sal, amplifying the oxygen losses due to solubility. We also observe an increase in mean water age in the S20-3 $\times$ -sal North Atlantic, which is due to the AMOC slowdown, and the intrusion of AABW into regions that were bathed by NADW in S20-1 $\times$ (Figure S8 in Supporting Information S1). This increase in water age contributes to increased TOU in the Atlantic basin interior in S20-3 $\times$ -sal. At depths below 3,000 m, TOU decreases correlate with younger water age due to extended northward penetration of AABW under high CO_2 . These changes in TOU modulate, but do not overcome, the deep oxygen concentration losses due to oxygen solubility.

Arctic waters exhibit widespread deoxygenation at high CO_2 levels in both configurations. This is especially pronounced in B21-2 $\times$ and B21-3 $\times$, which develop hypoxia across much of the Arctic Ocean below 250 m (Figure 5), driven by significantly higher water age (Figure 6). In contrast, the B21-1 $\times$ Arctic is well-ventilated. While S20-3 $\times$ and S20-3 $\times$ -sal also show reduced Arctic oxygenation, they do not exhibit widespread hypoxia. In contrast, these simulations exhibit an oxygen increase at higher CO_2 between 70°N and 75°N, driven by TOU decreases in the deep proto-Labrador Sea, which accompany changes to Labrador Sea deep convection (Hutchinson et al., 2025a). In each high- CO_2 case, Arctic deoxygenation is driven by a combination of reduced O_2^{pre} and increased TOU, with TOU impacts caused by increased respiration (Figure 7), and, in B21-3 $\times$, a significant contribution from increased water mass residence time (slower circulation; Figure 6, Figure S9 in Supporting Information S1).

3.3.3. Summary of Atlantic and Arctic Ocean Oxygenation

Our simulations show consistent declines in dissolved oxygen concentrations in both the Arctic and Atlantic Oceans under elevated CO_2 . This response is strongest under B21 paleogeographic boundary conditions, which exhibit hypoxic conditions in the North Atlantic between 300 and 2,000 m and throughout the Arctic below ~250 m (Section 3.3.2). In the B21 experiments, severe Atlantic deoxygenation arises from the combined effects of temperature-driven solubility loss and increased water mass residence times (allowing for more respiration) associated with a shutdown of the AMOC.

In contrast, the high- CO_2 S20 simulations maintain an active overturning circulation due to the more restricted Atlantic-Arctic exchange. Continued deep-water formation limits residence times and sustains ventilation of intermediate and deep waters, preventing the severe oxygen loss seen in B21. These results demonstrate that gateway geometry, through its control on freshwater exchange and overturning strength, exerts a strong influence on North Atlantic oxygen.

Despite the divergence in intermediate and deep ocean hypoxia, each high- CO_2 simulation exhibits less mean oxygen than its low- CO_2 reference, driven by solubility and ventilation decreases throughout the water column, and intensified by shoaled remineralization depth.

3.3.4. Pacific Ocean Oxygenation

Every high- CO_2 simulation exhibits greater Pacific mean oxygen than its $1\times\text{CO}_2$ reference (Table 2). This behavior diverges from trends of widespread deoxygenation at higher CO_2 in the Atlantic, and it is driven by oxygen concentration increases in specific regions and depth-ranges, which vary between each high- CO_2 simulation.

The Pacific is well oxygenated at high latitudes in all simulations, with much smaller hypoxic zones than in the Atlantic (Figure 8). In the Pacific sector of the Southern Ocean, oxygen decreases in all simulations with increased CO_2 (Figure 9). This is driven by a decrease in preformed O_2 , itself driven by decreased solubility. Oxygen at intermediate depths (down to about 1,000 m) also decreases in all simulations with increasing CO_2 , and this decrease is driven by changes in both preformed O_2 and TOU. In contrast, oxygen tends to increase substantially

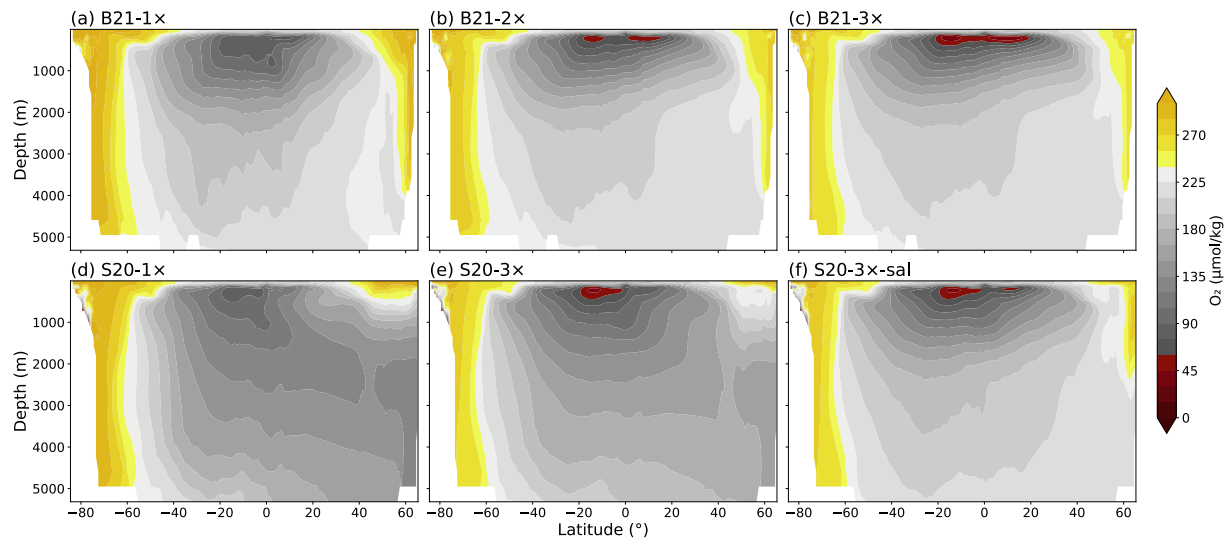


Figure 8. Pacific Ocean zonal average oxygen concentrations in $\mu\text{mol kg}^{-1}$ for (a–c) B21 and (d–f) S20 simulations.

below about 1,000 m in the tropical Pacific in all simulations, and in the North Pacific in the S20 configurations, driven by reduced TOU.

As was the case in the Atlantic, O_2^{pre} declines almost everywhere in the Pacific (Figure 9) due to globally decreased solubility. However, the decline is less pronounced for the S20 configurations owing to more vigorous NPDW formation at higher CO_2 that accesses deeper parts of the North Pacific (Figure 2, Figure S10 in Supporting Information S1). This counteracting circulation effect is most visible in the S20-3x-sal experiment, consistent with its stronger PMOC (Figure 9 and Figure S10 in Supporting Information S1).

Also like the Atlantic simulations, upper 1,000 m ocean oxygen decline is largely consistent throughout the high- CO_2 scenarios due to shoaled remineralization depth (Figure 7, Figure S10 in Supporting Information S1),

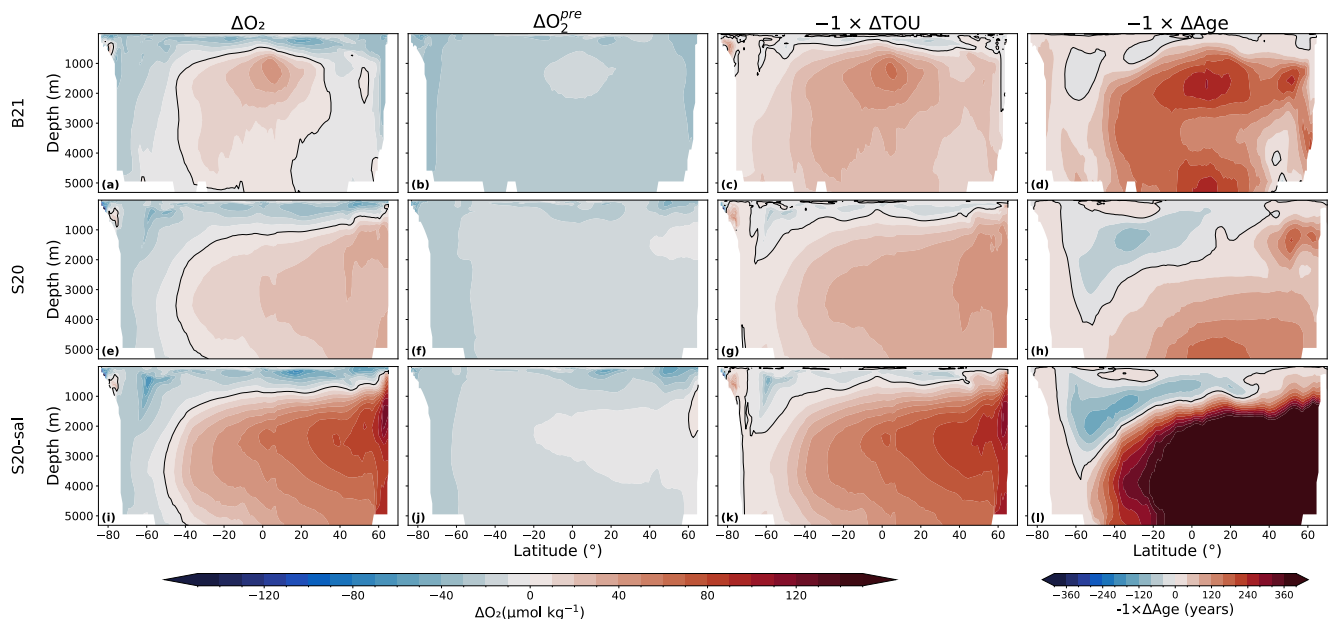


Figure 9. Difference in zonal mean Pacific Ocean (a, e, i) O_2 concentrations, (b, f, j) O_2^{pre} , (c, g, k) $-1 \times \text{TOU}$, in $\mu\text{mol kg}^{-1}$, and (d, h, l) $-1 \times \text{water age}$ in years for (a–d) B21-3x–B21-1x, (e–h) S20-3x–S20-1x, and (i–l) S20-3x-sal–S20-1x. We multiply TOU and water age by -1 to align the direction of change with their impact on oxygen concentration. The zero contour is plotted in black. B21-3x–B21-2x is shown in Figure S7 in Supporting Information S1.

reduced upper-ocean ventilation (Figure 9, Figure S10 in Supporting Information S1), and reduced solubility (Figure 9 and Figure S10 in Supporting Information S1). In the intermediate and deep ocean, TOU decreases throughout the Pacific are driven by decreased basin-interior remineralization and increased ventilation.

Water age increases between 50 and 70°S above 3,000 m in each simulation, which is due to AAIW formation weakening, allowing more time for respiration to act, removing more oxygen and increasing TOU. In addition, water age tends to increase in the shallow and intermediate depth middle latitudes (50°N–50°S) in both S20-3×CO₂ simulations, which contributes to stronger oxygen loss in the upper ocean.

TOU decreases in each high-CO₂ simulation in the Pacific below 1,000 m (Figure 9). These changes are primarily driven by a stronger PMOC and stronger AABW formation at high CO₂ (Figure 2) and associated with shorter water mass residence times. This effect is most pronounced in S20-3×-sal, which exhibits the most pronounced change in PMOC.

In addition, the TOU in B21-3× also decreases in the tropics at intermediate depths (1,000–3,000 m), driven by changes in the makeup and direction of the Atlantic-Pacific throughflow: in B21-1×, old NADW and TIW enter the Pacific via the open Central American Seaway (CAS), but their sources cease under 3×CO₂ (Section 3.3.2, Figure S11 in Supporting Information S1). This is discussed further in Section 3.3.7, which describes changes in the Eastern Tropical Pacific.

Remineralization changes further modulate deep-ocean TOU (below 1,000 m). In each simulation, basin-interior remineralization decreases at higher CO₂ (Figure 7) due to a temperature-driven shoaling of the remineralization depth, compounding the impact of enhanced ventilation in lowering TOU.

Localized increases in remineralization also occur. In all 3×CO₂ simulations we observe increased remineralization rates in the North Pacific, intruding to 4,000 m depth in B21-3×, 3,500 m depth in S20-3×-sal, and 3,000 m depth in S20-3× (Figure 7). This reflects greater transport of organic matter due to stronger NPDW formation. A similar change occurs in all three simulations at high southern latitudes. These remineralization increases do not, however, result in overall increased TOU, as they are counteracted by stronger bottom water formation and shorter residence times. Near the equator, we observe decreased remineralization throughout the water column, particularly above 300 m, which is present in all simulations but most pronounced in B21-3× (Figure S12 in Supporting Information S1). This decrease is caused by changes in the Eastern Tropical Pacific and will be analyzed in Section 3.3.7.

There is very limited proxy evidence for deep-ocean oxygenation available to validate oxygen in our simulations. Zhang et al. (2025) presented iron proxy reconstructions from the South China Sea at 2,000 m depth (ODP Site 1146; Figure 1). They found that this location was better oxygenated during the MMCT than during the MCO. Our B21-3× simulation is in agreement with these findings and presents less oxygen in this region than B21-1× (Figure S13 in Supporting Information S1). This oxygen decrease occurs despite a slight decrease in TOU at higher CO₂, and is driven by temperature increase. S20-3× and S20-3×-sal are in disagreement with Zhang et al. (2025). Both present increased oxygen concentrations at this depth despite decreased solubility, correlating with significantly reduced TOU due to more active NPDW ventilation (Figure S13 in Supporting Information S1).

3.3.5. Summary of Pacific Oxygenation

Pacific Ocean oxygen shows a more complex response to increasing CO₂ than the Atlantic Ocean. Although oxygen concentrations are impacted by basin-wide declines in solubility-driven O₂^{pre}, increased deep-ocean ventilation modulates overall oxygen concentration loss. Oxygen concentration increases in the deep ocean are TOU-driven and associated with stronger NPDW and AABW formation at higher CO₂, whereas TOU-driven oxygen declines at shallow and intermediate depth reflect weakened upper-ocean ventilation and shoaled remineralization.

3.3.6. Antarctic Shelf Oxygenation

The Southern Ocean (here defined as south of 60°S) is more oxygenated than other regions in all the B21 and S20 simulations (Table 2), and mean dissolved oxygen declines progressively with higher CO₂ (Sections 3.3.2 and 3.3.4). Here we analyze changes in dissolved oxygen directly over the Antarctic shelf, which exhibits changes in oxygen at higher CO₂ that cannot be fully explained by changes in solubility (Figures 6 and 9).

The Southern Ocean experiences significant shifts in convection sites under high CO₂ concentrations in our models (Figure S2 in Supporting Information S1). There is a southward shift in convection sites following sea ice retreat in all experiments. In addition, the mixed-layer deepens in the Pacific sector and shoals in the Indian sector of the Southern Ocean in all simulations. In the Atlantic sector, convection weakens overall, but there are some regions of enhanced convection in the southwestern proto-Weddell Sea (all simulations) and along the Antarctic coast east of the proto-Weddell Sea that reaches into the Indian Sector (B21-3×).

Changes in zonal mean oxygen over the Antarctic continental shelf reflect the shifts in convection. In the Atlantic, oxygen decreases over the shelves in all three simulations (Figure 6), due to a weakening and/or shift in convection and an associated increase in TOU. In the Pacific, all three simulations show a positive oxygen anomaly on the shelves (Figure 9) which is due to stronger convection increasing ventilation and hence decreasing TOU.

3.3.7. Eastern Tropical Pacific Oxygenation

The ETP (defined here as the Pacific east of 150°W, between 30°S and 30°N) is a region of the modern ocean with a persistent and expanding OMZ. However, available proxy evidence indicates that during the MCO, this region was well-oxygenated (Auderset et al., 2022; Hess et al., 2023).

In our simulations, the ETP exhibits higher mean oxygen concentrations with increasing CO₂ levels (Table 2), which occurs in spite of decreased zonal mean shallow (0–500 m) oxygen in each high-CO₂ simulation (Figure 9). The increase in water-column average oxygen is in part a result of increased deep-ocean ventilation and decreased remineralization at higher CO₂ in each configuration, analyzed in Section 3.3.4. However, in addition to the increase in deep oxygenation, B21-2× and B21-3× also show localized contraction of the OMZ between 500 and 1,500 m, which is not present in the S20 configurations (Figure 10, Figures S14–S16 in Supporting Information S1). This is driven by a combination of decreased productivity in the ETP and decreased oxygen utilization upstream of the ETP.

Hess et al. (2023) infer from foraminiferal I/Ca that the minimum oxygen concentration of the eastern tropical Pacific OMZ exceeded 100 μmol kg^{−1} during the MCO and declined to less than 7 μmol kg^{−1} during the cooler MMCT. Our simulations do not capture the extent of oxygenation inferred for the MCO and therefore do not capture the extent of deoxygenation inferred for the MMCT. This could be partially explained by the uncertainty of the proxy calibration (Zhou et al., 2022). However, the B21 simulations exhibit a decrease in both OMZ extent and intensity at higher CO₂ relative to B21-1×. S20 OMZs, by contrast, expand at higher CO₂. The B21 simulations capture the sign, but not the magnitude of the middle Miocene oxygen changes inferred by Hess et al. (2023).

Table 3 shows water properties averaged over hypoxic regions (O₂ ≤ 60 μmol kg^{−1}) in B21-1×. Compared to B21-1×, oxygen concentrations are higher in hypoxia-sensitive regions in B21-3×, both across the whole water column and above 1,000 m. This increase in oxygen occurs despite higher average water temperatures (Table 3). Additionally, PO₄ concentrations are lower, suggesting a decrease in remineralization and export production. Average water age is younger at all depths in B21-3×.

In the B21 simulations, changes in tracer properties within hypoxic regions correlate with shifts in water mass source contributions. Through the water column, the relative contribution of Southern Ocean source waters increases from 17.0% to 20.5% between B21-1× and B21-3×, alongside a substantial increase in North Pacific contributions (12.7%–23.1%), while North Atlantic contributions decline sharply (3.8% to <0.1%). Tethys-sourced water declines sharply as well (6.5%–0.2%).

A decrease in wind-driven upwelling in the higher CO₂ B21 simulations is the second driver of OMZ contraction. Figure 11 shows near-surface water age and surface wind direction anomalies for each 3×CO₂ simulation compared to their 1×CO₂ control run. Weaker southeasterly trade winds in B21-3× diminish upwelling, leading to younger surface waters, fewer available surface nutrients, less biological activity, less export production, and ultimately less oxygen consumption.

Using the set of change-attribution diagnostics described in Section 2.3.3, we are able to disentangle the contributions of changes in solubility, respiration, and circulation to ETP oxygen change for each high-CO₂ simulation (as per Equations 9 and 10; Pasquier et al., 2024). To visualize these contributions, we plot them as waterfall charts in Figure 12, which represents how the 1×CO₂ oxygen levels are incrementally increased or decreased by

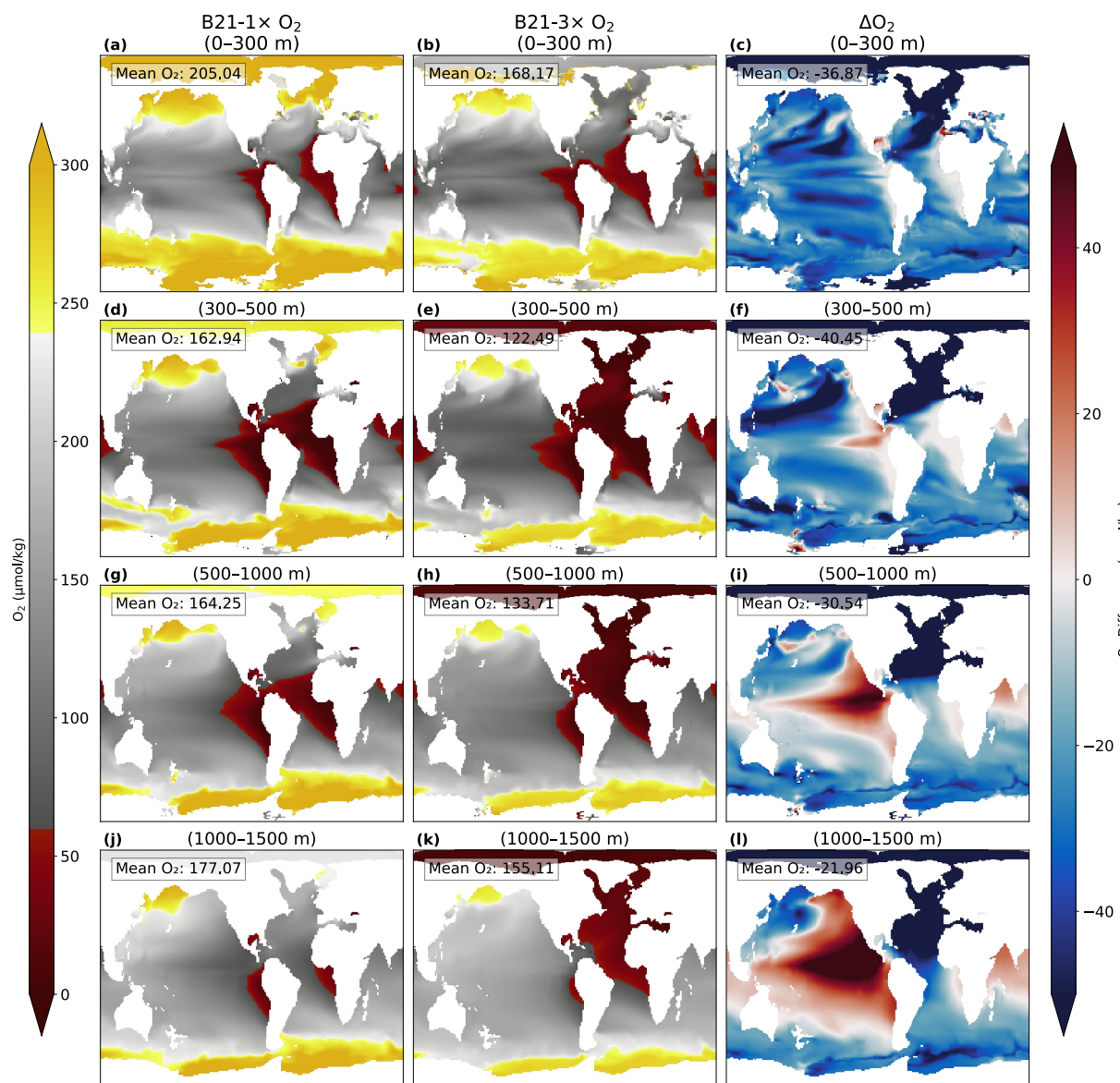


Figure 10. O_2 concentrations in $\mu\text{mol kg}^{-1}$ averaged between (a–c) 0–300 m, (d–f) 300–500 m, (g–i) 500–1,000 m, and (j–l) 1,000–1,500 m for (a, d, g, j) B21-1x, (b, e, h, k) B21-3x, and their difference, (c, f, i, l) ΔO_2 . (Similar figures for the other configurations can be found in Figures S14–S16 in Supporting Information S1).

Table 3

Volume-Weighted Mean Tracer Properties for Hypoxic Regions of the ETP as Defined Based on Oxygen Concentrations ($\leq 60 \mu\text{mol kg}^{-1}$) in B21-1x

Tracer	Depth range	B21-1x	B21-3x
O_2 ($\mu\text{mol kg}^{-1}$)	All depths	32.79	53.40
	<1,000 m	29.24	43.65
Temperature ($^{\circ}\text{C}$)	All depths	9.76	13.11
	<1,000 m	10.42	13.96
PO_4 ($\mu\text{mol kg}^{-1}$)	All depths	2.70	2.47
	<1,000 m	2.63	2.44
Water age (yr)	All depths	468.43	453.44
	<1,000 m	364.02	342.10

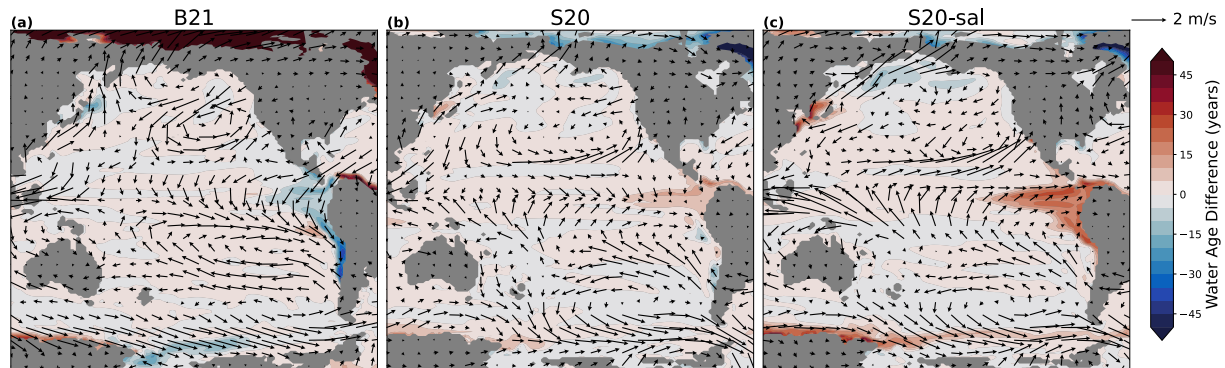


Figure 11. Modeled 50 m water age and wind direction difference for (a) B21-3x-B21-1x, (b) S20-3x-S20-1x, and (c) S20-3x-sal-S20-1x.

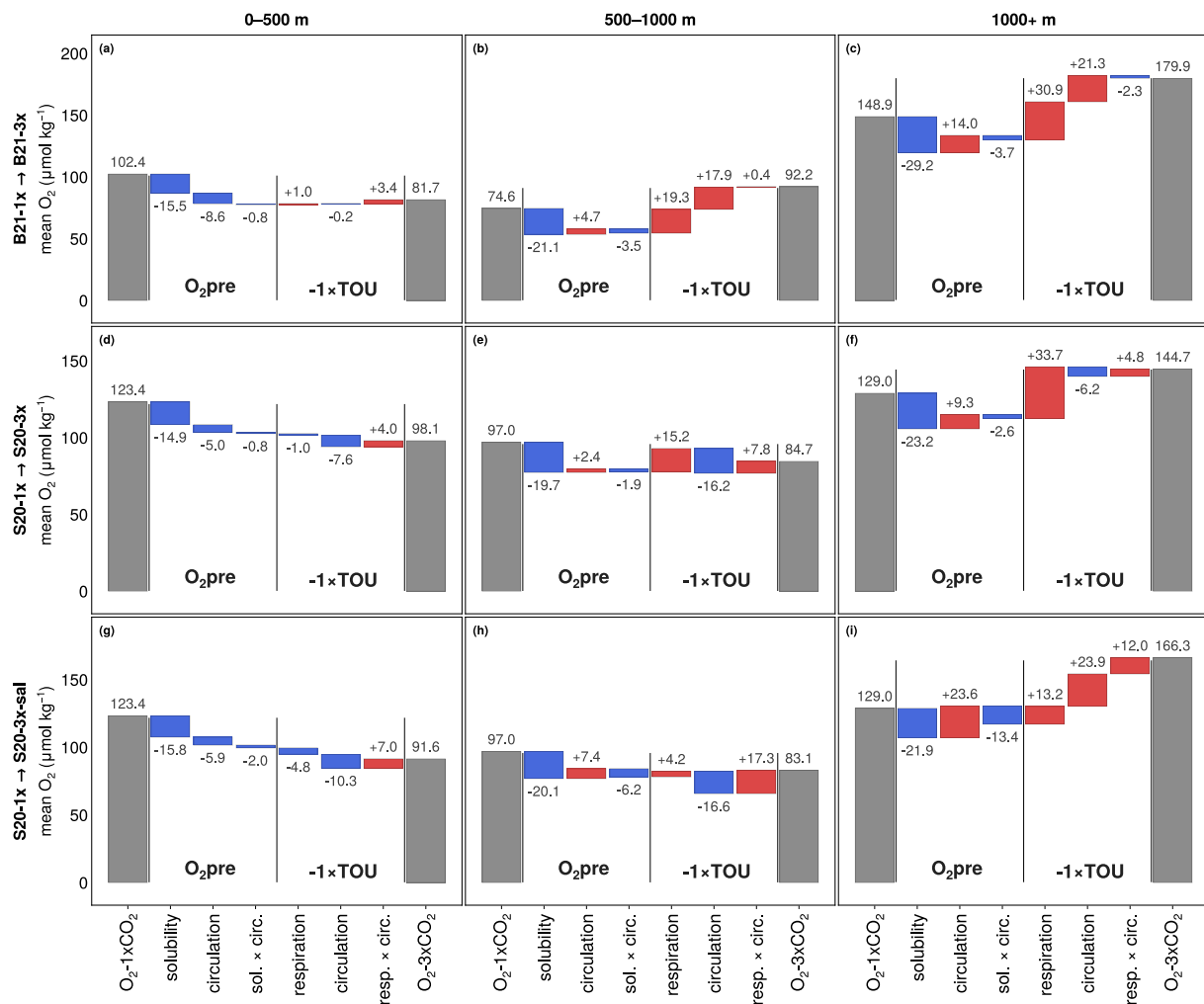


Figure 12. Waterfall decomposition of mean dissolved oxygen changes in $\mu\text{mol kg}^{-1}$ in the Eastern Tropical Pacific (ETP, Pacific Ocean east of 150°W and within 30°S – 30°N) relative to the $1\times\text{CO}_2$ reference state. Columns show averaged diagnostics for the (a, d, g) upper ocean (0–500 m), (b, e, h) intermediate depths (500–1,000 m), and (c, f, i) deep waters (1,000+ m). Rows correspond to high- CO_2 simulations for each paleogeographic configuration: (a–c) B21-3x, (d–f) S20-3x, and (g–i) S20-3x-sal, each evaluated relative to its own $1\times\text{CO}_2$ control. Red bars indicate positive contributions to ΔO_2 relative to the $1\times\text{CO}_2$ reference, whereas blue bars indicate negative contributions. Columns are read left to right, with each bar indicating the direction and magnitude of each driver's contribution to oxygen change. Gray bars indicate overall $1\times\text{CO}_2$ and $3\times\text{CO}_2$ oxygen concentrations. For example, a blue bar for O_2^{pre} “solubility” indicates relative loss in oxygen between $1\times$ and $3\times$ simulations due to decrease in average solubility. The partitions shown in these figures are expressed mathematically by Equations 9 and 10 (Section 2.3.3). We multiply TOU by -1 in order to visually represent oxygen concentration as the sum of O_2^{pre} and $-1\times\text{TOU}$.

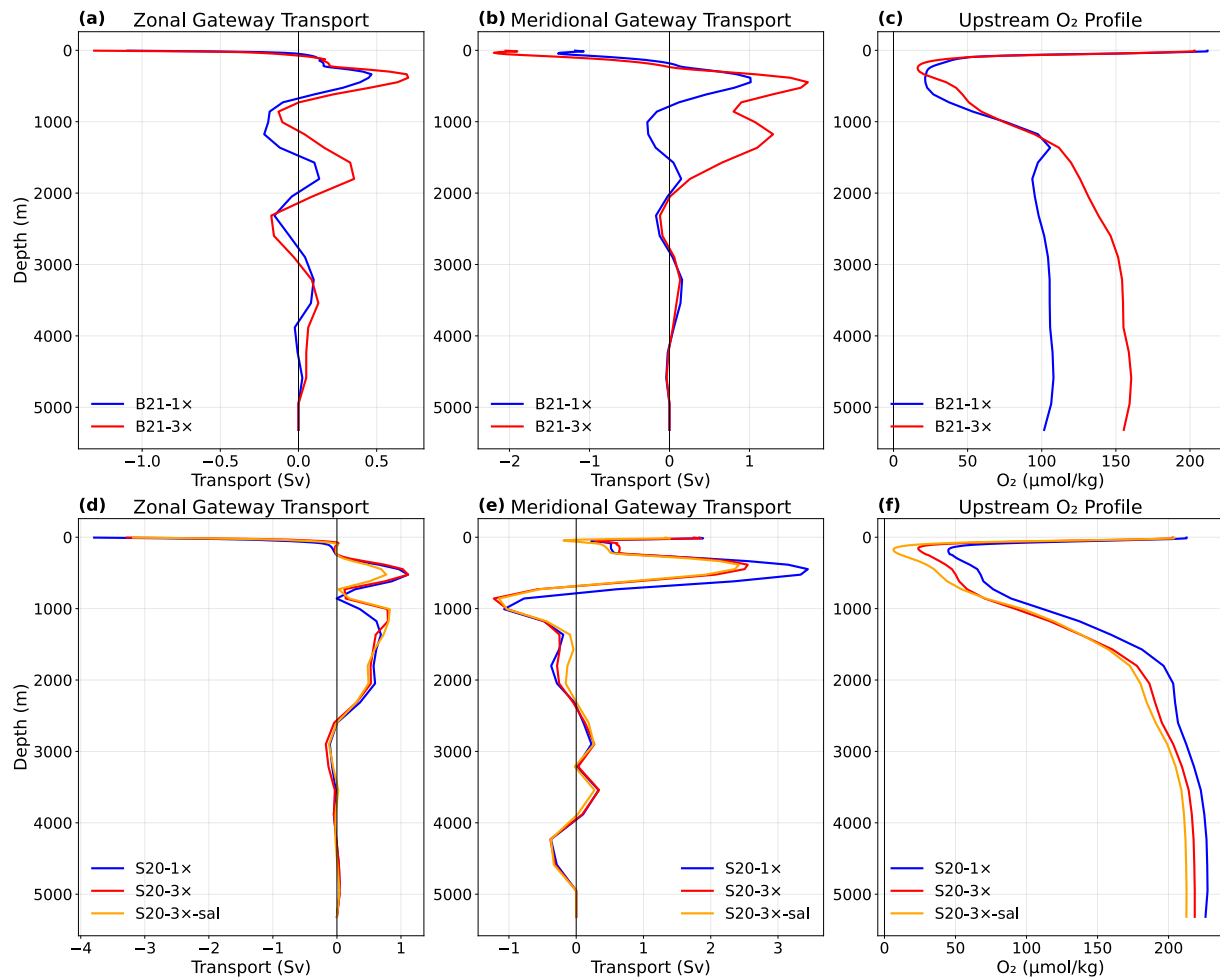


Figure 13. (a, d) Zonal and (b, e) meridional transport (Sv) through the Central American Seaway (CAS), as well as (c, f) O_2 concentration ($\mu\text{mol kg}^{-1}$) depth profiles averaged over the three cells “upstream” (on the Atlantic side) of the CAS for (a–c) B21 and (d–f) S20 simulations. Negative zonal and meridional transports indicate transport from the Atlantic into the Pacific ocean.

each contribution to produce the $3\times\text{CO}_2$ oxygen levels. Our decomposition confirms that in the B21-3 $\times$ ETP, between 500 and 1,000 m, a combination of respiration and circulation overcomes the loss in oxygen due to declining solubility (Figure 12b). Although the S20 simulations also exhibit some TOU decrease due to decreased respiration between 500 and 1,000 m (Figures 12e and 12h), it is to a lesser extent than in B21-3 $\times$ because it is counteracted by a circulation effect, leading to an overall expansion of OMZs. In B21-3 $\times$, decreased respiration and circulation changes both contribute to the gain in dissolved oxygen, indicating that the circulation-driven TOU decrease causes the difference between OMZ contraction or expansion in our simulations.

In the B21-1 $\times$ configuration, the open CAS and Atlantic-Pacific flow, particularly between 700 and 1,500 m, permits relatively old and deoxygenated NADW and TIW to flow into the ETP. At higher CO_2 , both of these water sources collapse, and the throughflow is replaced with younger waters (Figure S11 in Supporting Information S1).

The composition of waters in the ETP OMZ therefore changes in B21-3 $\times$. In this simulation, there is almost no TIW and NADW contributing to the ETP OMZ. In addition, the average transport across the CAS between 700 and 1,500 m depth reverses in B21-3 $\times$, from a southwestward to northeastward direction (from the Pacific to the Atlantic; Figures 13a and 13b). The resulting impact is that less O_2 is consumed on the way to the ETP from the Tethys and Atlantic, indicated by a decrease in circulation-driven TOU contributions from these regions. Figure 14 presents vertically integrated differences in the circulation-driven contribution to TOU change between B21-3 $\times$ and B21-1 $\times$. A water mass originating in the Tethys and North Atlantic giving rise to reduced TOU

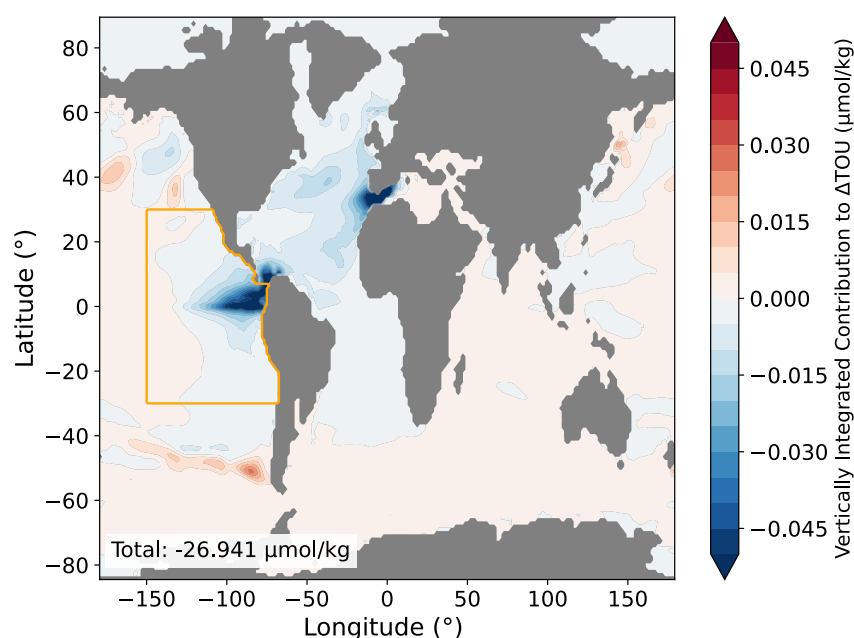


Figure 14. Vertically integrated upstream contribution to change in ETP (between 700 and 1,500 m) TOU (ΔTOU) due to circulation differences for B21-3 $\times$ –B21-1 $\times$ in $\mu\text{mol kg}^{-1}$. Upstream contribution to changes in circulation is as characterized in Equation 10 in Section 2.3.3. A grid cell's color indicates the difference in its contribution to the domain's (ETP 700–1,500 m) circulation-driven TOU (TOU–circulation) between high- CO_2 and reference simulation. A negative contribution in a grid cell therefore entails that water originating in that cell transports less TOU–circulation to the domain of interest in B21-3 $\times$ than in B21-1 $\times$. Orange outline indicates ETP region extent.

(negative changes, blue in the figure) indicates that these regions, across all depths, contribute less TOU to the domain of choice (in this case the ETP between 700 and 1,500 m) in B21-3 $\times$. The blue ΔTOU plume in the ETP itself reflects faster throughflow and therefore shorter residence times. Across the whole ocean, there is an increase of $26.9 \mu\text{mol kg}^{-1}$ in ETP O_2 concentrations between 700 and 1,500 m in B21-3 $\times$ due to circulation-driven TOU changes.

Similar changes are not observed in the S20 simulations, for which circulation-driven TOU increases in the ETP due to Tethys- and North Atlantic-sourced water (not shown), coinciding with smaller changes in CAS throughflow (Figures 13d and 13e). The difference between the B21 and S20 simulations is therefore driven primarily by a reorganization of Atlantic circulation between low- and high- CO_2 B21 states, which connect to the ETP in B21 through a deep CAS.

Additionally, unlike in the high- CO_2 B21 simulations, which see a decrease in upwelling-driven remineralization in the ETP, trade wind-driven southern hemisphere (SH) upwelling remains strong (S20-3 $\times$) or even intensifies (S20-3 $\times$ -sal) in the S20 simulations (Figure 11). We can explain these upwelling dynamics by analyzing atmospheric circulation, in particular, changes to the Hadley cell and Intertropical Convergence Zone (ITCZ).

In B21-3 $\times$, the Hadley circulation weakens and shifts southward, resulting in weaker SH trade winds and decreased upwelling (Figure 15, Figure S17 in Supporting Information S1). In contrast, we observe asymmetric changes in the Hadley circulation in S20-3 $\times$ -sal, where the southern branch of the Hadley cell strengthens and the northern branch weakens and shifts poleward, producing stronger SH trade winds and enhanced upwelling. We observe no obvious structural changes in S20-3 $\times$ (Figure 15, Figure S17 in Supporting Information S1). These trends in the trade winds are also apparent in the longitudinally integrated zonal wind, which shows weakened and southward shifted trade winds in B21-3 $\times$ in the SH (Figure 16). These SH easterlies strengthen and move north in the high- CO_2 S20 simulations. This dynamic reflects an asymmetric poleward shift of the ITCZ, which reduces zonal pressure gradients and weakens trade winds in the hemisphere toward which it shifts. Figure S18 in Supporting Information S1 shows equator-normalized mean precipitation in each simulation, indicating a southward shifted ITCZ in B21-3 $\times$ and much smaller changes to ITCZ location in S20.

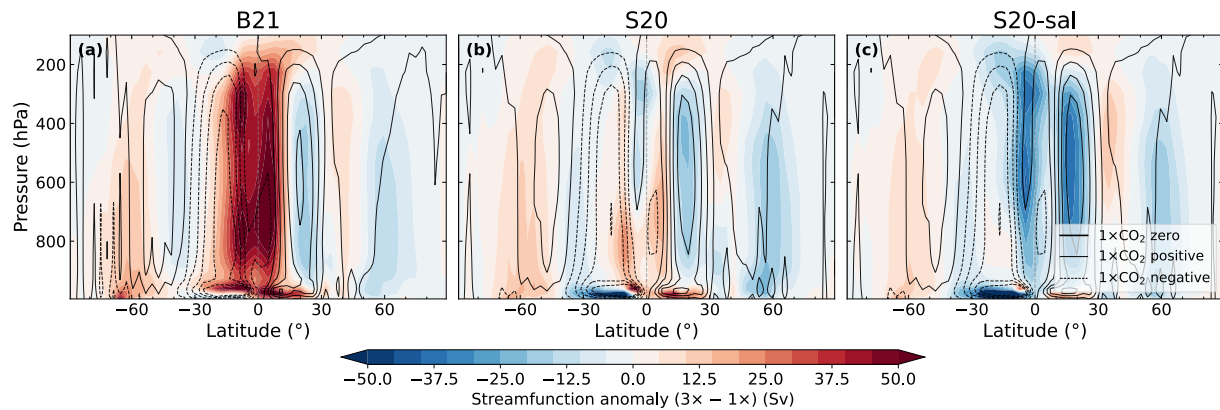


Figure 15. Atmospheric meridional streamfunction (Ψ , Sv) anomaly ($3\times - 1\times\text{CO}_2$) for (a) B21, (b) S20, and (c) S20-sal, with $1\times\text{CO}_2$ simulation contours overlaid. Figure S17 in Supporting Information S1 shows full fields.

These changes in Hadley circulation and zonal wind strength are consistent with surface air temperature (SAT) gradient anomalies, which indicate subdued mid-latitude northern hemisphere (NH) warming in B21- $3\times$ and stronger mid-latitude NH warming in S20- $3\times$ -sal (Figure S19 in Supporting Information S1). These SAT gradient differences are driven by changes in large-scale oceanic heat transport. AMOC collapse in B21- $3\times$ reduces poleward heat transport in the NH, resulting in a relatively warmer SH and a southward shift of the Hadley cell. This response is consistent with prior model studies that link AMOC strength to interhemispheric temperature gradients and ITCZ position (e.g., McGee et al., 2018; Schneider et al., 2014). In contrast, the activation of the PMOC in S20- $3\times$ -sal increases northward heat transport in the Pacific, driving relatively stronger warming in the NH (Figure S19 in Supporting Information S1), while there are no major circulation reorganizations between S20- $3\times$ and S20- $1\times$. These interhemispheric SAT gradient differences are consistent with the observed Hadley circulation responses in each simulation, with a southward ITCZ shift in B21- $3\times$, a northward ITCZ shift in S20- $3\times$ -sal, and no shift in S20- $3\times$.

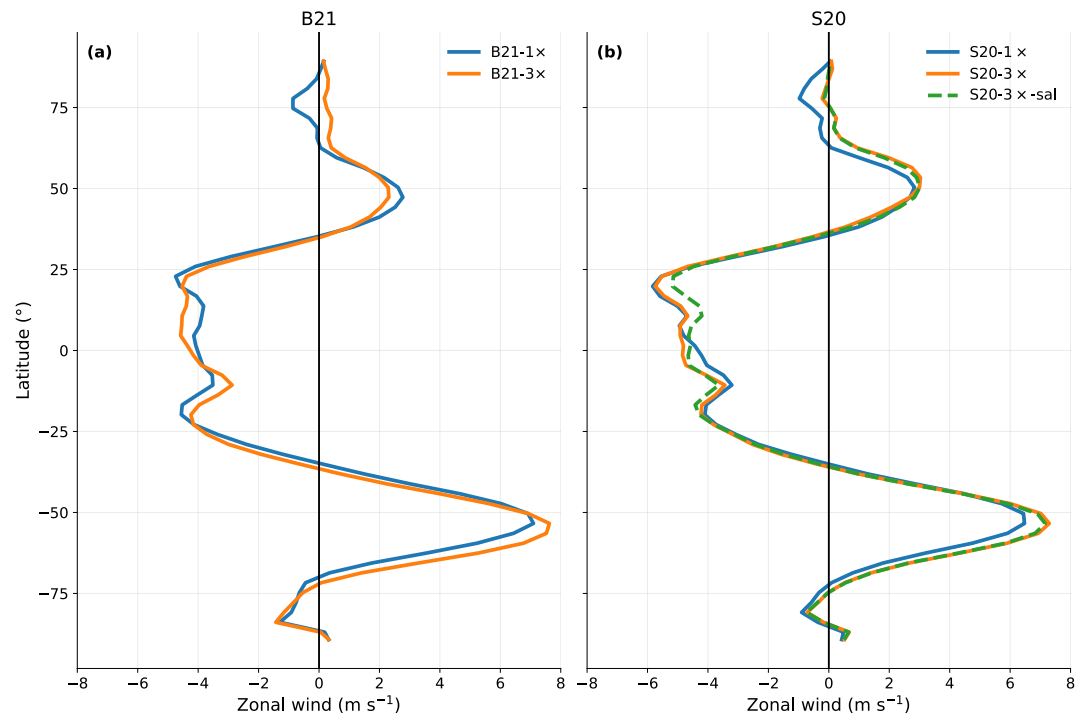


Figure 16. Mean zonal wind speeds in m/s over the Pacific basin (here 100°E – 60°W) for (a) B21 and (b) S20 simulations. Values greater than 0 indicate westerly winds.

3.3.8. Summary of Eastern Tropical Pacific Oxygenation

The enhanced ETP oxygenation in the high- CO_2 B21 simulations aligns with Miocene proxy evidence. Multi-proxy records from the ETP indicate well-oxygenated conditions during the warm MCO ($>100 \mu\text{mol kg}^{-1}$) followed by deoxygenation during MMCT cooling (Auderset et al., 2022; Hess et al., 2023). The B21 suite reproduces the sign of this observed trend: a better-oxygenated ETP under MCO-like CO_2 (B21-2 $\times$ and B21-3 $\times$) and deoxygenation under cooler MMCT-analog conditions (B21-1 $\times$).

Our analysis of interrelated biogeochemical, ocean, and atmospheric dynamics provides three-dimensional insight into the MCO oxygenation described by Hess et al. (2023) and Auderset et al. (2022). Contrary to proxy evidence, we do not observe a complete absence of OMZs in the high- CO_2 ETP, but we do find agreement with general trends in our B21 simulations.

The contraction of the ETP OMZ in the B21-3 $\times$ simulation is driven by a combination of weaker upwelling and younger water ages owing to decreased Atlantic-Pacific transport of oxygen-depleted waters through the CAS. Weaker upwelling is linked to weaker trade winds due to a weaker and southward-shifted Hadley circulation, while the water age decrease is driven by reduced transport of oxygen-depleted NADW and TIW through the CAS due to the shutdown of their formation at high CO_2 . Neither of the high- CO_2 S20 simulations exhibits an OMZ contraction, although at depths greater than 1,500 m they, like the B21 simulations, both exhibit enhanced oxygenation. We illustrate the processes driving OMZ contraction in B21-3 $\times$ in schematic form (Figure 17).

4. Discussion

Our model simulations offer insight into the three-dimensional physical and biogeochemical dynamics operating in a range of plausible middle Miocene steady states. Below, we discuss these results in the context of existing modeling and proxy literature, address the implications and limitations of our model-proxy comparison, and explore how Miocene oxygen dynamics relate to modern and future warming.

4.1. Climate Drivers of Simulated OMZs

Across all high- CO_2 simulations, we find consistent upper ocean deoxygenation driven by reduced preformed oxygen concentrations, shoaling of the remineralization depth, and enhanced stratification, regardless of paleogeographic configuration. However, gateway geometry significantly impacts the oxygenation of the Atlantic ocean interior and the extent of OMZs in the ETP.

Our results are broadly consistent with other model simulations of long-term oxygen responses to warming. Pasquier et al. (2024) found widespread deoxygenation in a perpetually warm idealized ocean using the PCO2 model, and our simulations generally reproduce this signal at upper and intermediate depths, as well as throughout the water column in the Atlantic, with the ETP under B21 boundary conditions a regional exception. Our deep Pacific results also share similarities with the work of Yamamoto et al. (2015) and Frölicher et al. (2020), who found that sustained warming eventually reinvigorates deep overturning and oxygenates deep waters in transient simulations. Stronger NPDW and AABW formation in our high- CO_2 simulations reduces deep Pacific TOU, representing a steady-state analog of this ventilation recovery. The tropical Pacific's divergence from the global signal of deoxygenation in our simulations also echoes past studies. Fu et al. (2018) found that tropical OMZ expansion under RCP8.5 may reverse as early as 2300, driven by declining tropical export production and ventilation changes, and Busecke et al. (2022) found that, in CMIP6 projections, low-oxygen Pacific waters expand while the OMZ core contracts. Our B21 simulations also exhibit a non-uniform response in the ETP, with shallow oxygen declining at high CO_2 while the OMZ between 500 and 1,500 m contracts.

Our simulations also qualitatively agree with those of Burke et al. (2026), who found that an open CAS promotes Pacific oxygenation and Atlantic deoxygenation, with nutrient advection from the Pacific to the Atlantic driving expanded Atlantic hypoxia regardless of atmospheric CO_2 . Burke et al. (2026) also reported increased Atlantic hypoxia associated with AMOC collapse, which they induced via salinity flux adjustment. In our simulations, a deeper CAS in B21 corresponds to a more oxygenated Pacific and a less oxygenated Atlantic than in S20 at all CO_2 levels. The nutrient-advection mechanism is CO_2 -independent, and so does not explain the ETP OMZ contraction at high CO_2 , which in our simulations is driven by changes in large-scale ocean and atmospheric circulation.

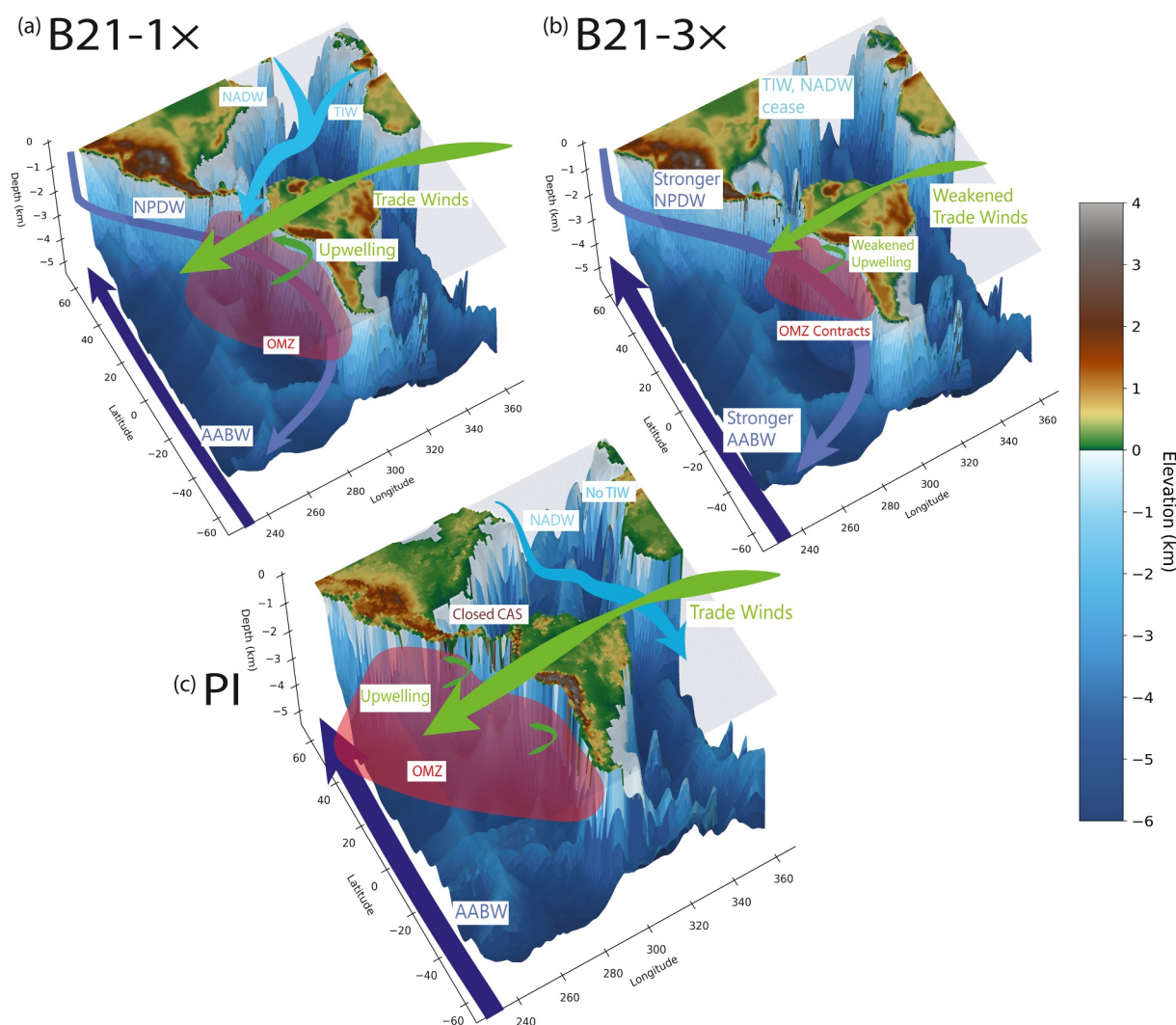


Figure 17. Schematic representation of the drivers contributing to the ETP OMZ in (a) B21-1x, (b) B21-3x, and (c) pre-industrial. In B21-3x, TIW and NADW shut down and the transport of these water masses through the CAS ceases. NPDW formation strengthens, and AABW formation strengthens moderately. Trade winds weaken in the southern hemisphere, leading to weaker upwelling. The pre-industrial control features an active AMOC, closed CAS, and more extensive ETP OMZ. Red shading in the ocean shows OMZ extent.

The atmospheric circulation dynamics contributing to the OMZ contraction in the ETP in our B21 simulations are also consistent with prior modeling studies. The connection between MOC state and ITCZ position is well-documented: in the modern context, ITCZ position has been linked to MOC heat transport, which causes hemispheric temperature asymmetry. The persistence of the AMOC thus displaces the ITCZ north of the equator (Marshall et al., 2014). As seen in our B21 simulations, AMOC weakening has been associated with a southward shift of the Hadley cell and/or weakening of SH trade winds in previous studies (McGee et al., 2018; Orihuela-Pinto, England, & Taschetto, 2022; Orihuela-Pinto, Santoso, et al., 2022; Schneider et al., 2014). Furthermore, Pontes et al. (2022) showed that a northward ITCZ shift can drive intensified Pacific trade winds, analogous to the Hadley cell shift and strengthened SH trade winds we see in our S20-3x-sal simulation.

4.2. Model—Proxy Comparison

No single simulation reproduces all aspects of the available proxy record, but the B21 configuration performs best overall. These simulations capture the water mass characteristics (Holbourn et al., 2024), the carbonate chemistry trends (Kender et al., 2014), and the sign of ETP OMZ change between MCO and MMCT analogs (Auderset et al., 2022; Hess et al., 2023). The S20 configuration, by contrast, produces an unchanged or expanded ETP OMZ

at high CO₂, opposite to proxy evidence. This disagreement is informative: the S20-3x-sal simulation, which develops an active PMOC like the B21 simulations, confirms that Pacific overturning intensification alone is insufficient, and that a more ventilated deep Pacific does not guarantee OMZ contraction. Changes in trade winds, upwelling, and remineralization are also not sufficient on their own to lead to OMZ contraction in our simulations. Respiration rates in the ETP OMZ (500–1,000 m) weaken in all simulations, which is consistent with potentially shallower remineralization depth and less carbon burial as observed in Li et al. (2023). Only the B21 paleogeographic configuration, with its wider Atlantic-Arctic gateway leading to an AMOC collapse at high CO₂, reproduces the ETP OMZ trend recorded in proxies. This suggests that the Atlantic-Arctic gateway geometry is critical for Miocene ocean circulation and biogeochemistry.

A notable feature of our B21 high-CO₂ simulations is that Atlantic hypoxia and ETP oxygenation are not independent, as both are impacted by the same circulation reorganization. NADW and TIW collapse weakens ventilation of the Atlantic while simultaneously eliminating the supply of old, oxygen-depleted NADW and TIW to the ETP through the CAS. We note that ETP oxygen proxies from periods with an open CAS should be interpreted with awareness that they may reflect Atlantic circulation changes in addition to changes in local upwelling and productivity.

Our results highlight several cautions for interpreting Miocene oxygen proxies. The strong sensitivity of OMZ extent to gateway geometry means that single-site records may not reflect basin-wide trends, particularly in the ETP where Atlantic circulation signals can be transmitted through the CAS. Our simulations identify the North Atlantic-Arctic gateway width and CAS state (open or closed) as first-order controls on regional oxygen, yet both remain subject to significant paleogeographic uncertainty. Better constraints on these gateways would improve future modeling efforts.

4.3. Caveats and Limitations

Our study has limitations due to the parent GFDL simulations, PCO2 model, and Miocene data scarcity. The GFDL CM2.1 model underestimates polar amplification for both modern warming and past warm climates (e.g., Hutchinson et al., 2025a; Winton, 2011). This bias affects high-latitude temperatures and oxygen solubility, with consequences for global deep-ocean oxygen via solubility and deep-water convection. This bias could also accentuate differences in atmospheric circulation: a reduced meridional temperature gradient could weaken SH upwelling and drive more widespread OMZ contraction in the ETP, whereas stronger polar warming could reorganize the ocean circulation with less predictable regional oxygen consequences.

Our simulations are steady-state equilibrium simulations under fixed orbital parameters, paleogeography, and greenhouse gas concentrations. They are not transient simulations with varying orbital parameters and therefore cannot shed light on the orbital-driven processes leading to multi-millennial variability recorded in high-resolution proxies.

PCO2 omits several key biogeochemical complexities, such as phytoplankton community diversity, the role of zooplankton grazing, explicit representations of the silicate, carbonate, and iron cycles, and processes specific to OMZs, such as denitrification (Chien et al., 2023; Kvale et al., 2019, 2021; Nissen et al., 2021). The PCO2 model only includes a single nutrient, phosphorus, and therefore cannot explicitly represent the effects of denitrification. However, when oxygen concentrations fall below a threshold of 5 μmol kg⁻¹, remineralization continues but no longer impacts oxygen concentration, which is intended to approximate anaerobic remineralization. Our results are thus likely to overestimate remineralization in suboxic waters and do not include any potential regional or global effects due to changes in the chemical composition of macro-nutrient pools.

Another limitation of our model stems from an absence of middle Miocene-era biogeochemical constraints. Specifically, in the absence of global constraints for the Miocene, we assume that global mean ocean alkalinity and total nutrient content were equal to today. There is, however, regional evidence that deep Pacific [PO₄³⁻] increased through the late Miocene toward modern values (Wang et al., 2022). If this signal reflects a global shift, middle Miocene nutrient inventories may have been lower than assumed here, and we would therefore overestimate productivity and oxygen depletion. The global alkalinity inventory is likewise poorly constrained. Long-term Cenozoic reconstructions indicate DIC and total alkalinity were broadly similar to their modern values, but resolving changes on finer timescales is difficult (Zeebe & Tyrrell, 2019). Alkalinity affects our results only indirectly through carbonate chemistry. Future research could quantify the sensitivity of ocean oxygen to these

assumptions by running a computationally efficient model with fixed circulation across a range of global mean nutrient and alkalinity values (e.g., Behrooz et al., 2024; Monteiro et al., 2012).

Because of these limitations, we interpret the simulated oxygen fields in terms of their overall structure and responses to perturbed boundary conditions, rather than treating them as hindcasts of middle Miocene oxygen concentrations.

4.4. Steady-State Biogeochemistry Models for Paleoclimate Applications

Our study demonstrates both strengths and weaknesses of steady-state biogeochemical solvers, such as PCO₂, as applied to paleoclimate investigations. Although this tool cannot resolve transient dynamics or internal variability by design, it is appropriate for assessing snapshots of equilibrium climates, especially climates that are very different from present day. Such solvers find the steady-state of a set of biogeochemical tracers at low computational cost compared to standard time-stepping models. Running on eight cores, the solver converged within 12 hr of wall-clock time (96 CPU-hours). In contrast, time-stepping the GFDL CM2.1 model with online biogeochemistry for the required time to reach equilibration (~5,000 years) and at the same resolution requires approximately 3,250 hr (~4.5 months) of wall clock time on 96 cores (312,000 CPU-hours, which is over 3,000× the cost of PCO₂).

Building the transport matrix for the PCO₂ solver requires only standard model outputs as input fields (Chamberlain et al., 2019). That is, if archived standard outputs are available, running the parent coupled model is not required. The standard outputs needed for the transport matrix are water-mass transports and mixed-layer depths, while age tracers are used to validate the imposed diffusivity, as described by Chamberlain et al. (2019). PCO₂ additionally requires air-sea fluxes, which in turn require temperature, salinity, surface winds, sea-ice cover, and shortwave radiation.

The greatest limitation of the PCO₂ model for paleoclimate research lies in the difficulties of direct model-proxy comparison. PCO₂ currently lacks isotopic partitioning and an explicitly resolved nitrogen cycle. These limitations make quantitative comparisons difficult. Finally, PCO₂'s lack of explicit seasonal variability means that intra-annual dynamics are averaged out, posing problems for identifying the seasonal controls of OMZs (e.g., the Arabian Sea OMZ is highly dependent on seasonal monsoonal dynamics). It is possible to encode climatological seasonal dynamics within the transport-matrix framework, for example by using monthly circulation matrices and boundary conditions (e.g., Huang et al., 2021; Pasquier et al., 2025), although this increases computational expense substantially.

5. Conclusions

Proxy evidence from Auderset et al. (2022) and Hess et al. (2023) indicates a contraction of the ETP OMZ during the high-CO₂ Miocene Climatic Optimum, which runs counter to current trends of transient OMZ expansion under increasing atmospheric CO₂. Our analysis offers a mechanistic explanation for the changes suggested by paleoclimate proxies, simulating steady-state middle Miocene ocean biogeochemistry under a range of CO₂ levels and two distinct paleogeographic reconstructions. Weaker upwelling and altered flow through the CAS produce a contracted OMZ at higher CO₂. This ETP OMZ contraction occurs concurrently with, and is partly driven by the same mechanism as, widespread hypoxia throughout the Atlantic and Arctic. Cessation of NADW and TIW formation cuts off modes of Atlantic and Arctic ventilation, while reducing the supply of old, oxygen-depleted water flowing into the ETP through the CAS; it also causes a southward shift of the Hadley cell and weakens trade-wind-driven upwelling, reducing oxygen consumption in the ETP OMZ. This circulation reorganization produces varied oxygen responses in different ocean basins, underscoring the spatial complexity of the oxygen response to long-term warming in steady state.

Generally, we find that under middle Miocene boundary conditions, oxygen declines at higher CO₂ throughout most of the global ocean; however, ocean oxygen responses to warming are complex and sensitive to large-scale modes of ocean overturning, gateway geometry, and atmospheric dynamics, none of which are static over deep time. This sensitivity is further underscored by the fact that ETP OMZ contraction emerges only under the B21 paleogeographic configuration and not S20, highlighting how gateway geometry can determine whether a basin experiences oxygenation or deoxygenation under otherwise identical forcing.

The boundary conditions responsible for ETP oxygenation at higher CO₂ concentrations in our simulations include an open CAS, limiting direct extrapolation from MCO evidence to future warming. However, upper ocean deoxygenation, driven by reduced preformed oxygen, shoaling remineralization, and increased stratification, is consistent across all high-CO₂ configurations and is relevant to future warming. Future progress will require better constraints on Miocene gateway geometry and multi-basin proxy compilations capable of testing the Atlantic-Pacific oxygen relationships exhibited by our simulations.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

A repository of relevant code and data for running the PCO2 model with GFDL CM2.1 matrices and Miocene boundary conditions is available via a Zenodo repository (Berg et al., 2026). Climate and ocean circulation outputs from the GFDL CM2.1 simulations can also be found on Zenodo (Hutchinson et al., 2025b).

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